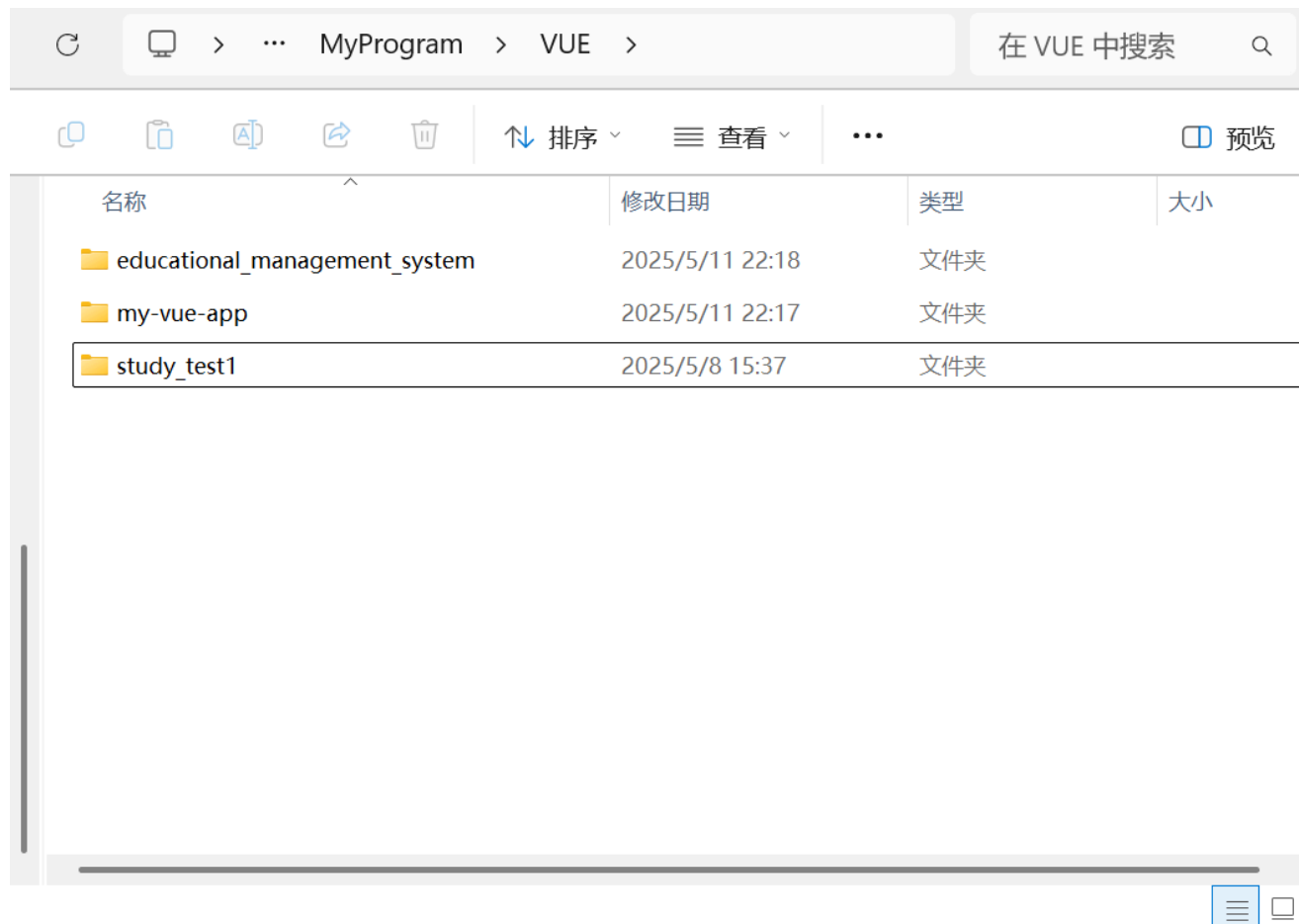


如何使用 Vite 创建一个 Vue 项目

本项目基于 Vue 3+TS，使用 pnpm。故需先下载 pnpm，教程见网络。也可替换 pnpm 为 npm。若替换为 npm，则部分安装依赖语句需要更换语法。

1.使用 Vite 创建项目

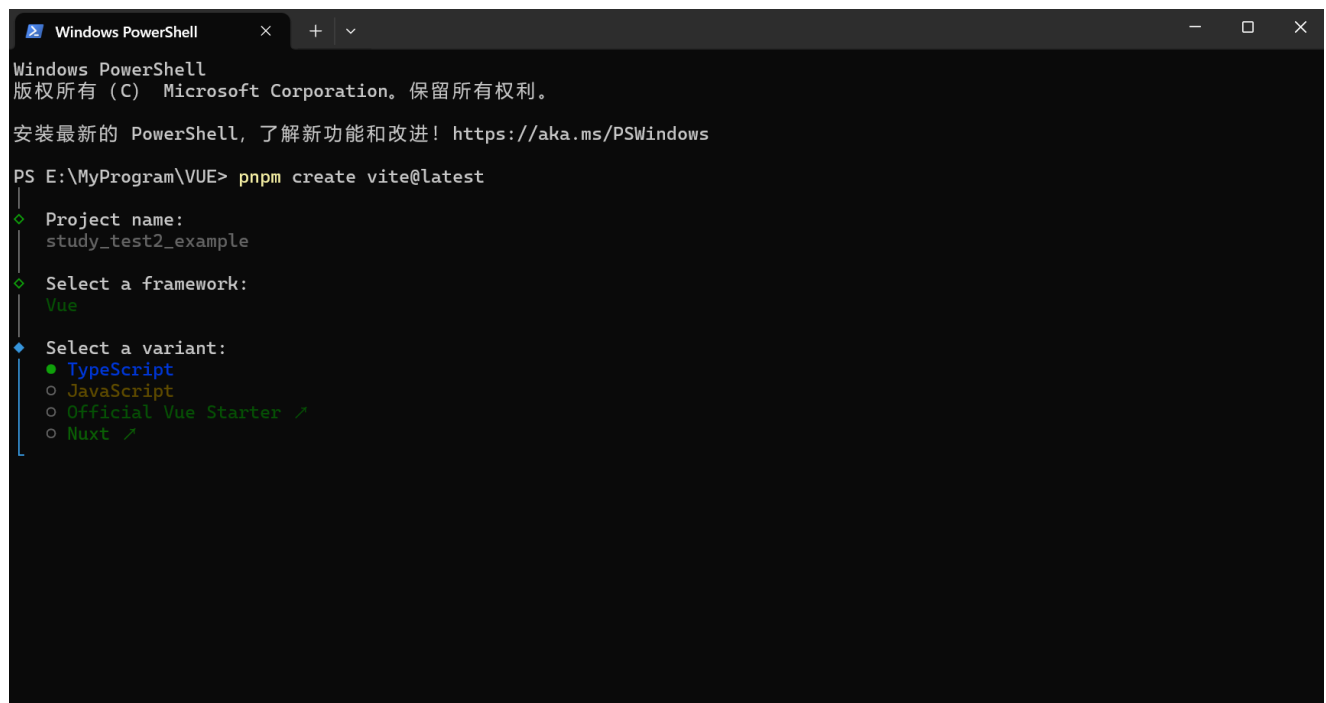
打开需要创建项目的大文件夹。

打开终端，输入 `pnpm create vite@latest`，根据提示输入项目名称，选择框架。



再选择语言。

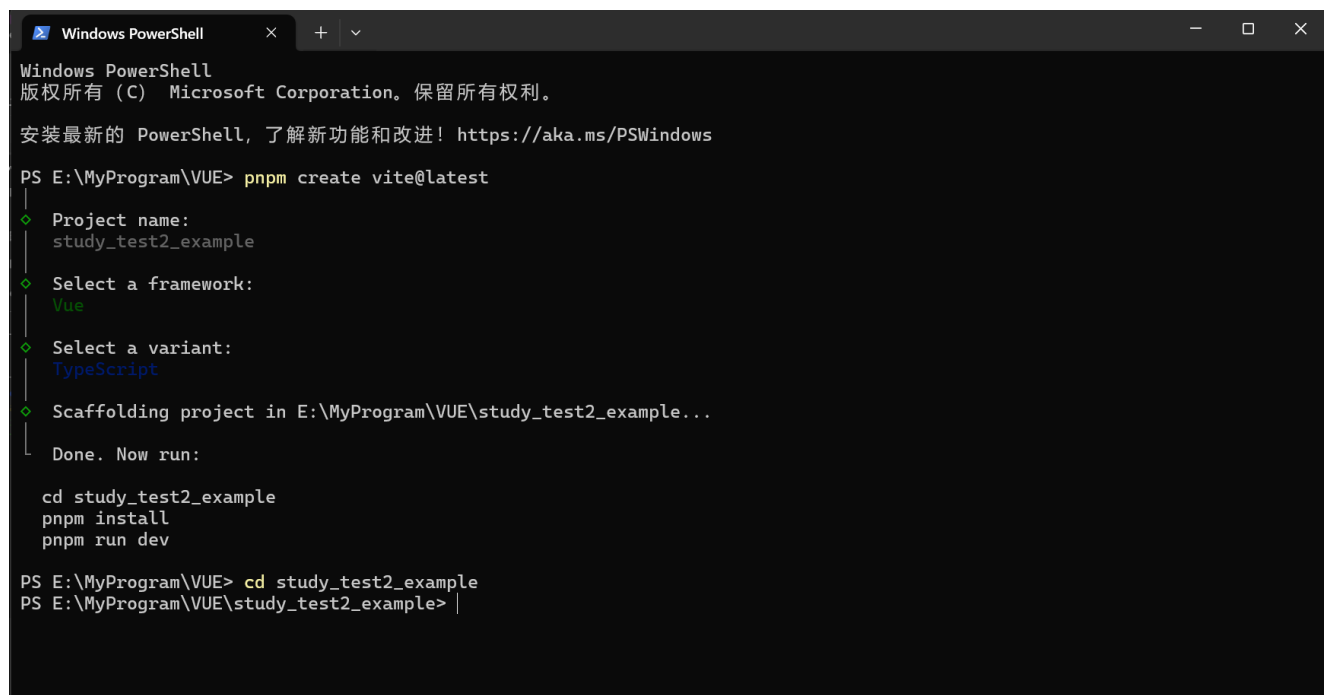


```
Windows PowerShell
版权所有 (C) Microsoft Corporation。保留所有权利。

安装最新的 PowerShell，了解新功能和改进！ https://aka.ms/PSWindows

PS E:\MyProgram\VUE> pnpm create vite@latest
|
|   Project name:
|   study_test2_example
|
|   Select a framework:
|   Vue
|
|   Select a variant:
|   • TypeScript
|   ○ JavaScript
|   ○ Official Vue Starter ↗
|   ○ Nuxt ↗
|
```

根据提示语切换到刚创建的文件夹，并进行初始化。



```
Windows PowerShell
版权所有 (C) Microsoft Corporation。保留所有权利。

安装最新的 PowerShell，了解新功能和改进！ https://aka.ms/PSWindows

PS E:\MyProgram\VUE> pnpm create vite@latest
|
|   Project name:
|   study_test2_example
|
|   Select a framework:
|   Vue
|
|   Select a variant:
|   TypeScript
|
|   Scaffolding project in E:\MyProgram\VUE\study_test2_example...
|
|   Done. Now run:
|
|   cd study_test2_example
|   pnpm install
|   pnpm run dev
|
PS E:\MyProgram\VUE> cd study_test2_example
PS E:\MyProgram\VUE\study_test2_example> |
```

```
Windows PowerShell
PS E:\MyProgram\VUE\study_test2_example> pnpm install

Update available! 10.14.0 → 10.16.1.
Changelog: https://pnpm.io/v/10.16.1
To update, run: pnpm self-update

Packages: +48
Progress: resolved 94, reused 11, downloaded 37, added 48, done

dependencies:
+ vue 3.5.21

devDependencies:
+ @vitejs/plugin-vue 6.0.1
+ @vue/tsconfig 0.7.0 (0.8.1 is available)
+ typescript 5.8.3 (5.9.2 is available)
+ vite 7.1.5
+ vue-tsc 3.0.7

Warning
Ignored build scripts: esbuild.
Run "pnpm approve-builds" to pick which dependencies should be allowed to run scripts.
```

```
PS E:\MyProgram\VUE\study_test2_example> pnpm run dev

> study_test2_example@0.0.0 dev E:\MyProgram\VUE\study_test2_example
> vite

You are using Node.js 22.6.0. Vite requires Node.js version 20.19+ or 22.12+. Please upgrade your Node.js version.

VITE v7.1.5 ready in 596 ms

→ Local: http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help
```

点击连接，显示页面如下。



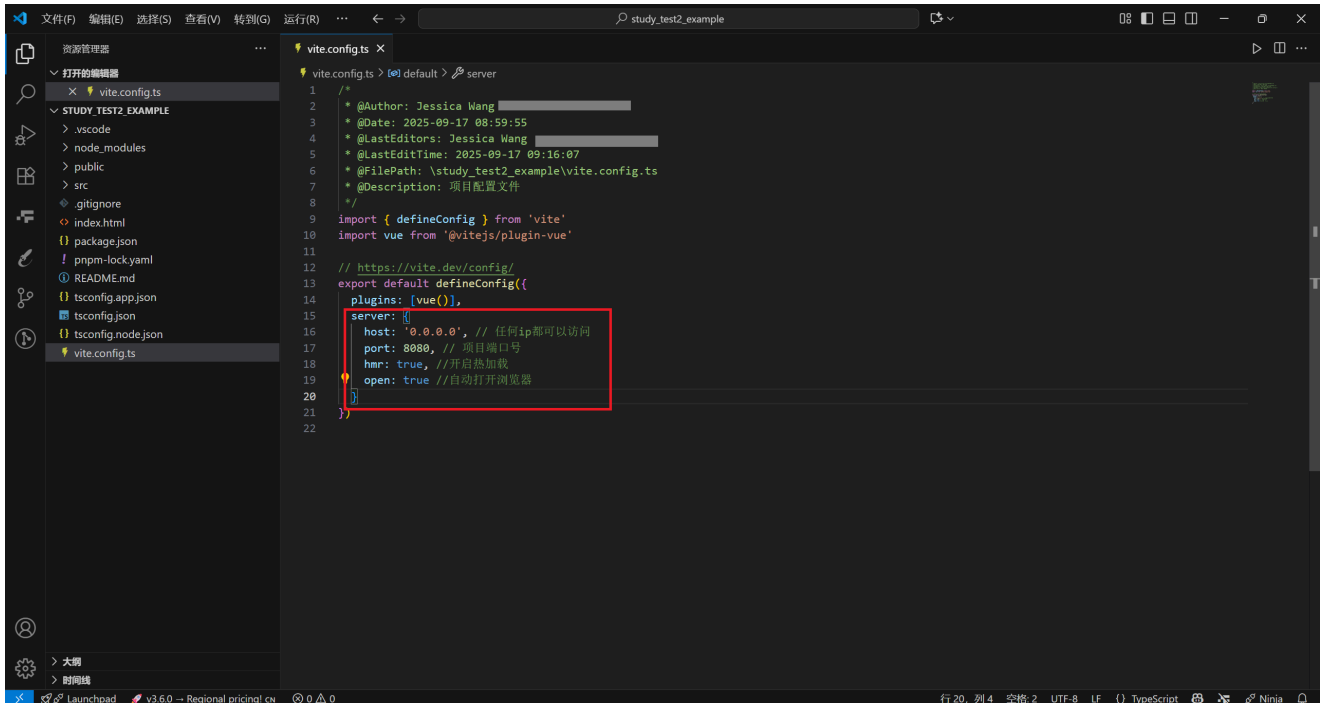
2.安装路由

打开刚才创建的文件，本文使用 VSCode 打开。

2.1 配置别名

2.1.1 vite.config.js 配置文件，添加如下配置

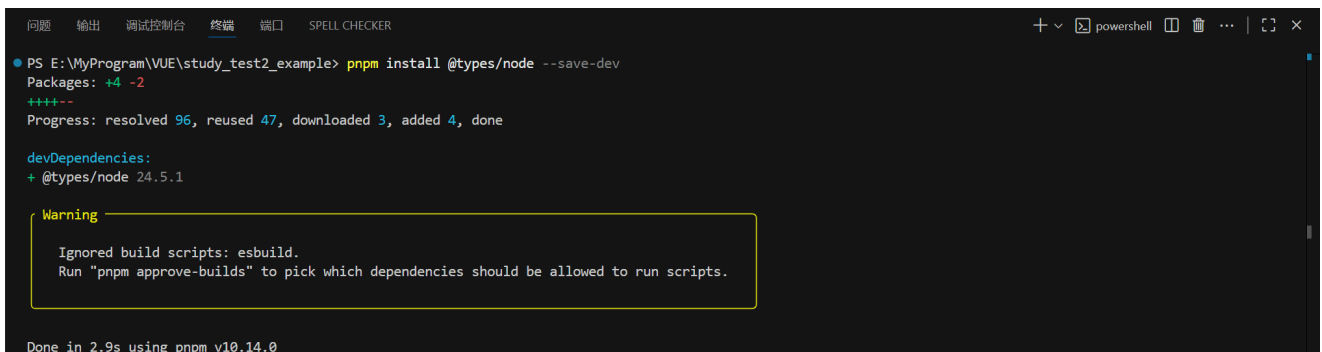
```
1  server: {
2    host: '0.0.0.0', // 任何ip都可以访问
3    port: 8080, // 项目端口号
4    hmr: true, //开启热加载
5    open: true, //自动打开浏览器
6  },
```



2.1.2 vite 配置别名

在项目的终端安装必须的依赖。

```
1  npm install @types/node --save-dev
```



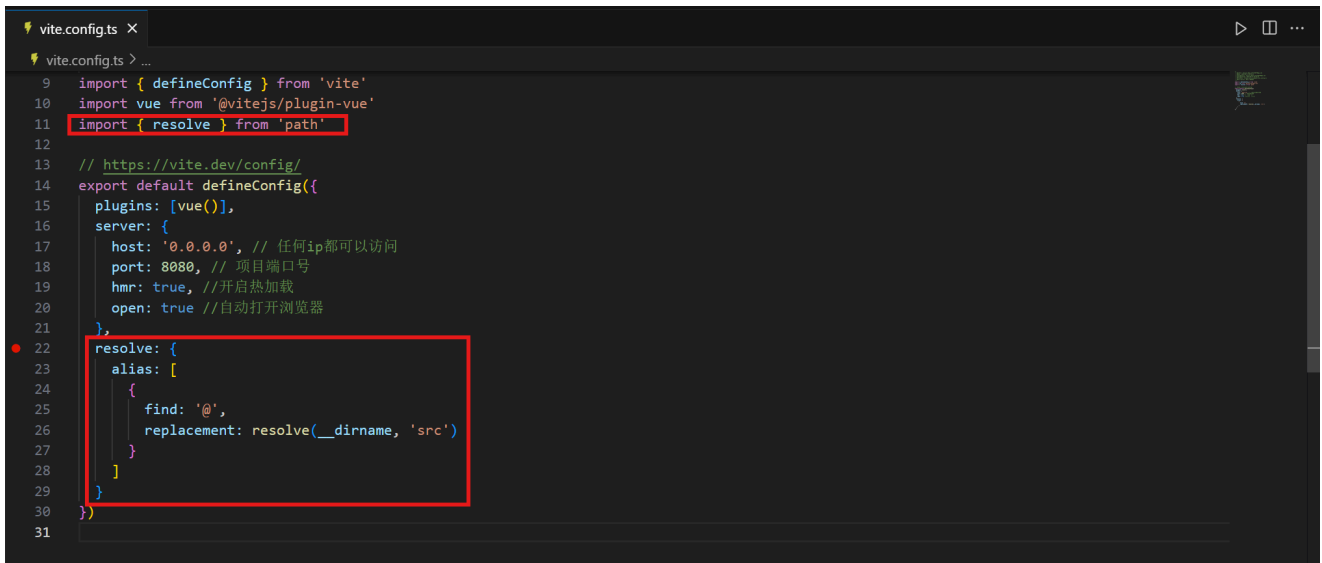
在 vite.config.ts 中添加代码。

```
1  import { resolve } from 'path'
2
3  resolve: {
4    alias: [
```

```

5      {
6        find: '@',
7        replacement: resolve(__dirname, 'src')
8      }
9    ]
10  },

```



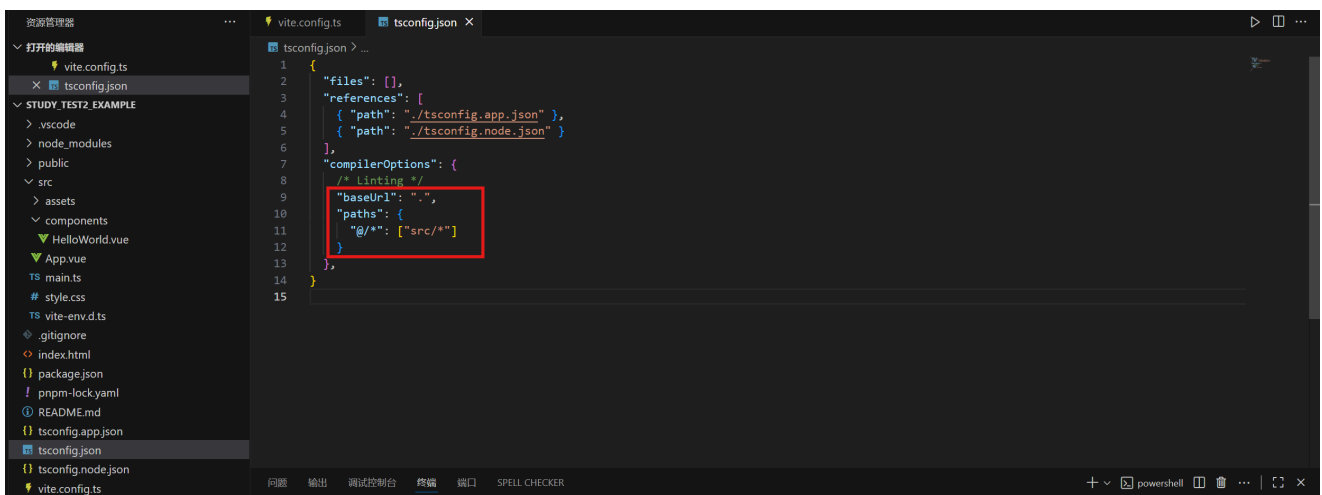
2.1.3 添加 baseUrl 和 paths

`tsconfig.json` 里面添加如下代码。

```

1  "compilerOptions": {
2    "baseUrl": ".",
3    "paths": {
4      "@/*": ["src/*"]
5    },
6  },

```

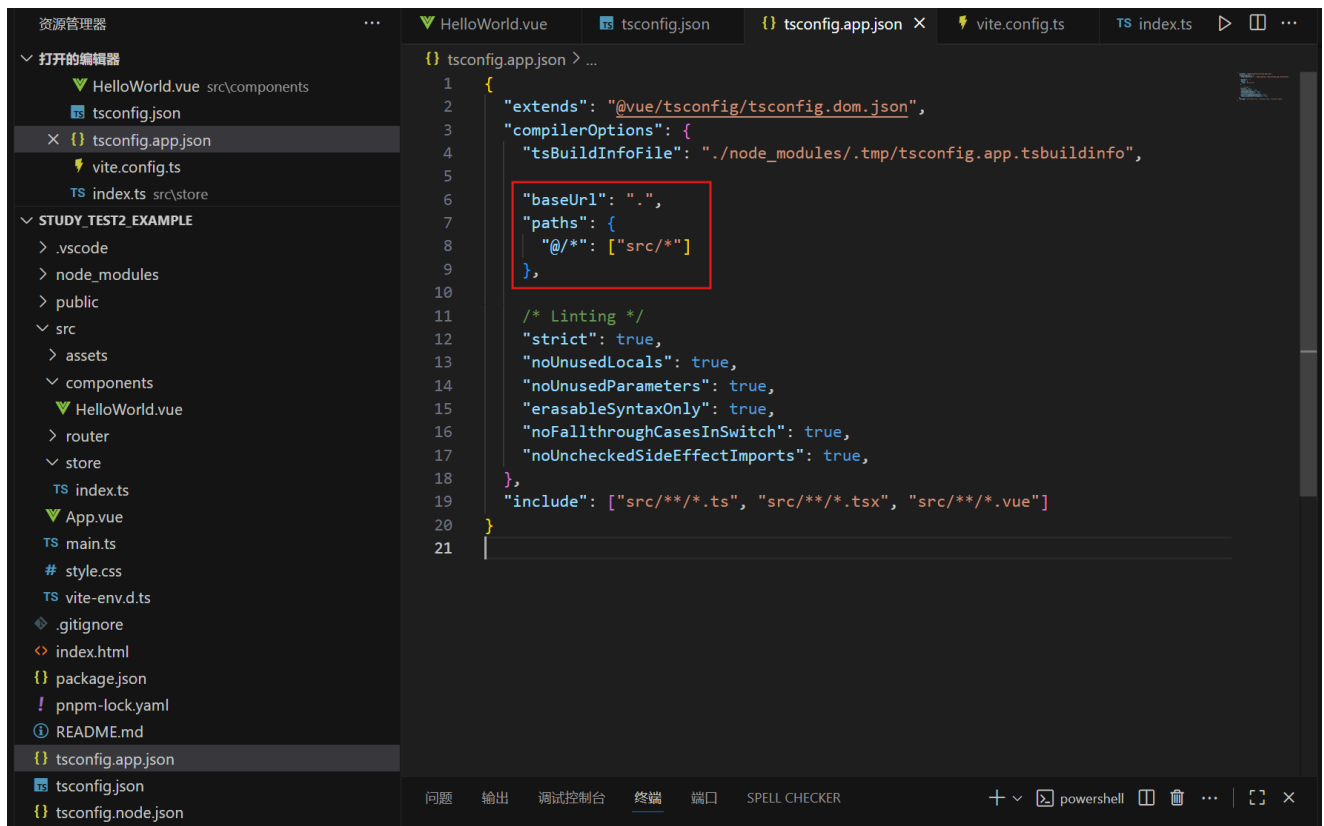


若 `tsconfig.json` 文件中存在 `{ "path": "./tsconfig.app.json" }`，则在 `tsconfig.app.json` 文件中也添加如下代码。

```

1  "baseUrl": ".",
2  "paths": {
3    "@/*": ["src/*"]
4  },

```



拓展：了解 tsconfig.app.json 和 tsconfig.node.json 的不同

特性	tsconfig.app.json	tsconfig.node.json
目标环境	浏览器环境	Node.js 环境
编译目标	ES2020 / ESNEXT	ES2022
模块系统	ES Module	CommonJS / ES Module
用途	前端应用代码	Vite 配置和构建脚本
包含文件	src/**/*.ts	vite.config.ts, package.json 等
输出目录	dist	无（通常不输出）

1. tsconfig.app.json - 前端应用配置

特点：针对浏览器环境优化，支持 Vue 文件，严格类型检查。

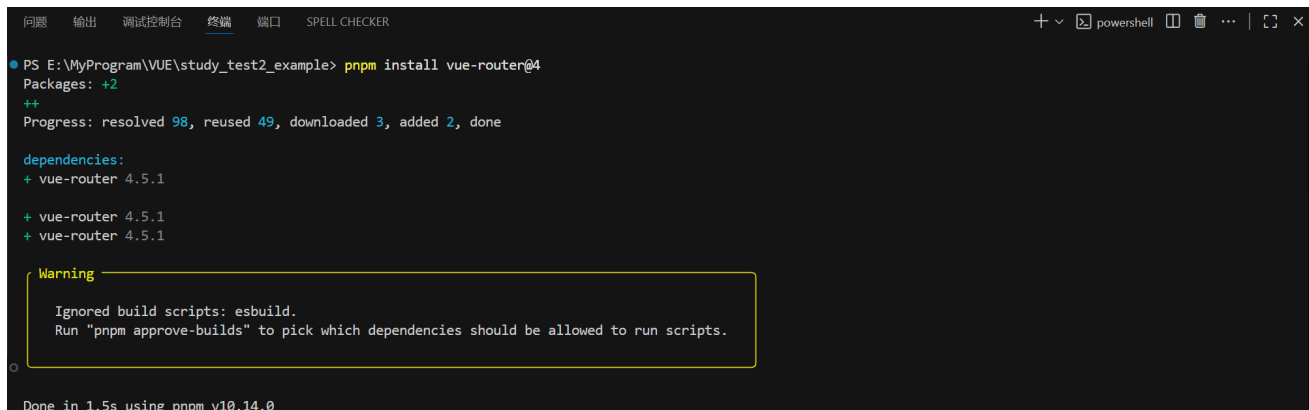
2. tsconfig.node.json - Node.js 环境配置

特点：针对 Node.js 环境，用于类型检查构建配置文件。

2.2 安装路由

2.2.1 安装依赖

```
1 pnpm install vue-router@4
2 # pnpm add vue-router@4
```

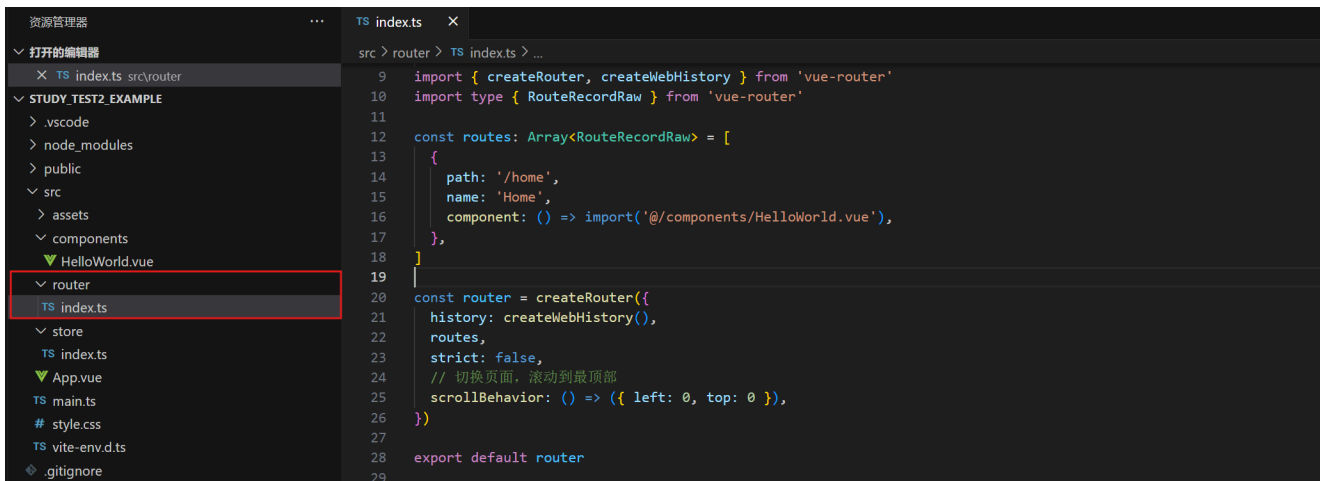


A terminal window showing the execution of the command `pnpm install vue-router@4`. The output indicates that 2 packages were added, with a progress bar showing 98 resolved, 49 reused, 3 downloaded, and 2 added. A warning box is displayed, stating: "Warning: Ignored build scripts: esbuild. Run 'pnpm approve-builds' to pick which dependencies should be allowed to run scripts." The terminal also shows the dependencies list, which includes `vue-router 4.5.1` three times. The command was completed in 1.5s using pnpm v10.14.0.

2.2.2 新建路由文件

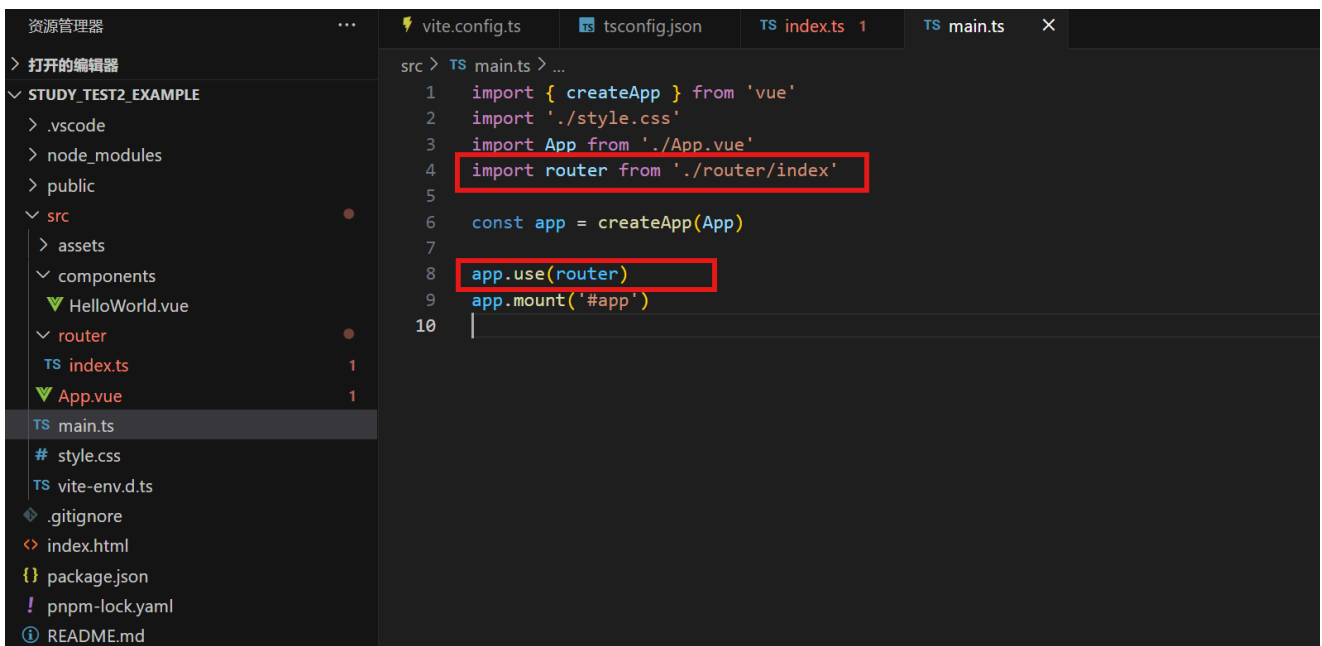
在 src 目录下新建 router 文件夹，然后新建 `index.ts` 文件

```
1 import { createRouter, createWebHistory } from 'vue-router'
2 import type { RouteRecordRaw } from 'vue-router'
3 import Layout from '@/components/HelloWorld.vue'
4
5 const routes: Array<RouteRecordRaw> = [
6   {
7     path: '/home',
8     name: 'Home',
9     component: Layout,
10   },
11 ]
12
13 const router = createRouter({
14   history: createWebHistory(),
15   routes,
16   strict: false,
17   // 切换页面，滚动到最顶部
18   scrollBehavior: () => ({ left: 0, top: 0 }),
19 })
20
21 export default router
```



2.2.3 在 main.ts 里面引入路由

```
1 import { createApp } from 'vue'
2 import './style.css'
3 import App from './App.vue'
4 import router from './router/index'
5
6 const app = createApp(App)
7
8 app.use(router)
9 app.mount('#app')
```

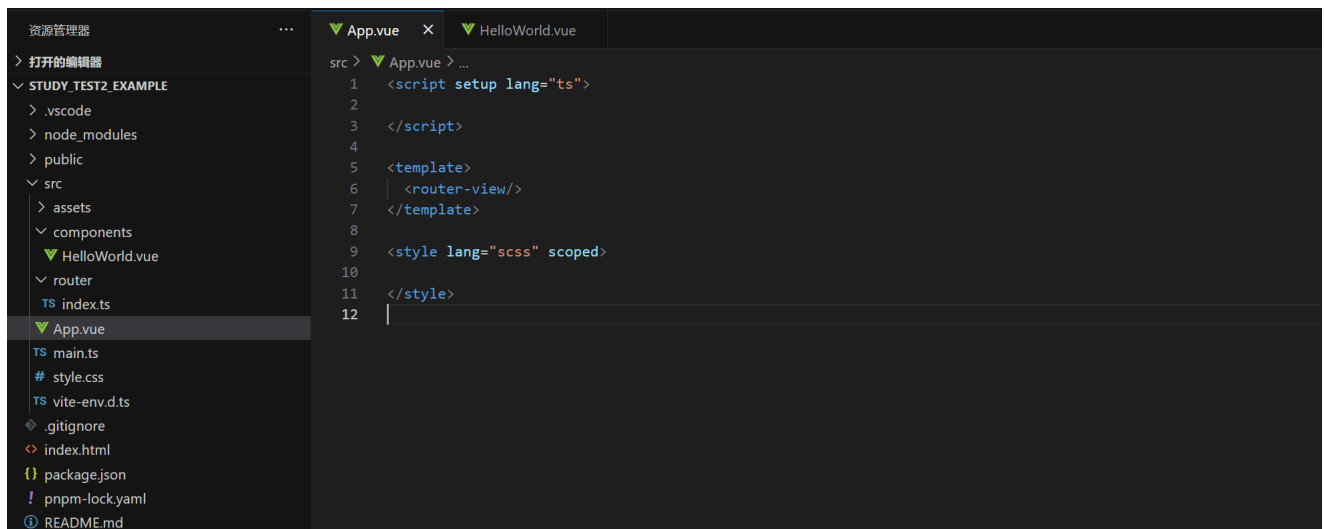


2.2.4 修改 App.vue 和 HelloWorld.vue

修改 App.vue 内代码为

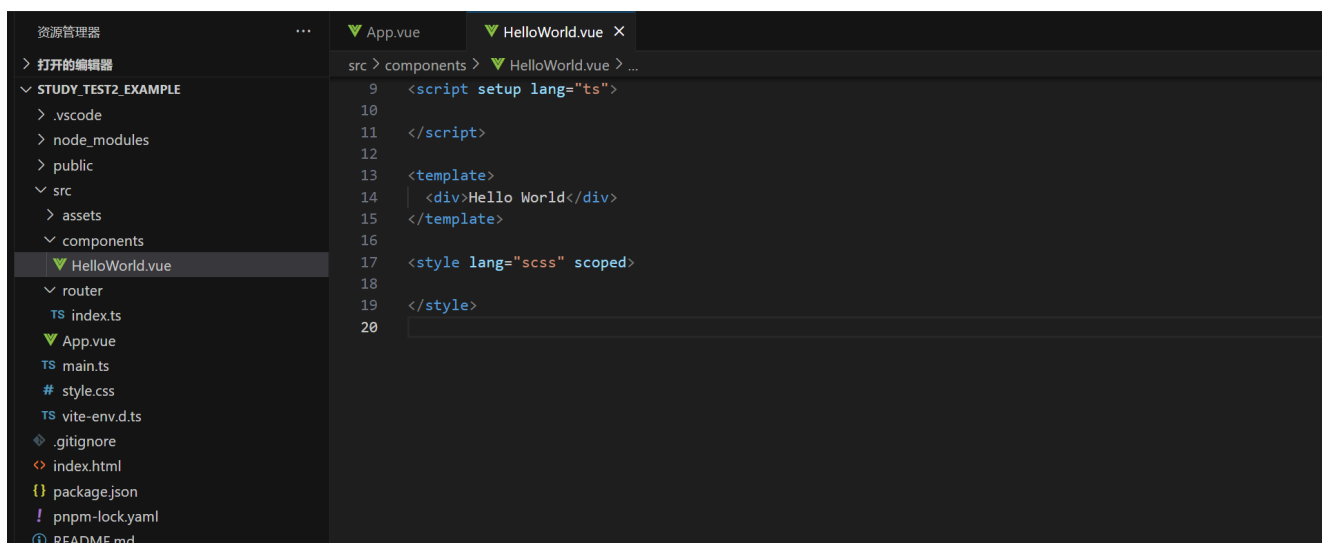
```
1 <script setup lang="ts">
2
3 </script>
4
```

```
5   <template>
6     <router-view/>
7 </template>
8
9   <style lang="scss" scoped>
10
11 </style>
```



修改 HelloWorld.vue 内代码为

```
1   <script setup lang="ts">
2
3   </script>
4
5   <template>
6     <div>Hello World</div>
7   </template>
8
9   <style lang="scss" scoped>
10
11 </style>
```



2.2.5 运行项目，检验结果

在终端输入。

```
1  pnpm run dev
```

```
PS E:\MyProgram\VUE\study_test2_example> pnpm run dev

> study_test2_example@0.0.0 dev E:\MyProgram\VUE\study_test2_example
> vite

X vite v7.1.5 is not yet supported in the Community edition of Console Ninja.
We are working hard on it for you https://tinyurl.com/3h9mtwra.

Estimated release dates:
- Community users: around 8th November, 2025 (subject to team availability)
- PRO users:      priority access is available now

You are using Node.js 22.6.0. Vite requires Node.js version 20.19+ or 22.12+. Please upgrade your Node.js version.
10:44:06 [vite] (client) Re-optimizing dependencies because lockfile has changed

VITE v7.1.5 ready in 4692 ms

→ Local:   http://localhost:8080/
→ Network: http://10.97.183.121:8080/
→ press h + enter to show help
```

若无法正确查看页面，可在终端输入 `pnpm install vite.config.ts` 重启后则可正常显示。

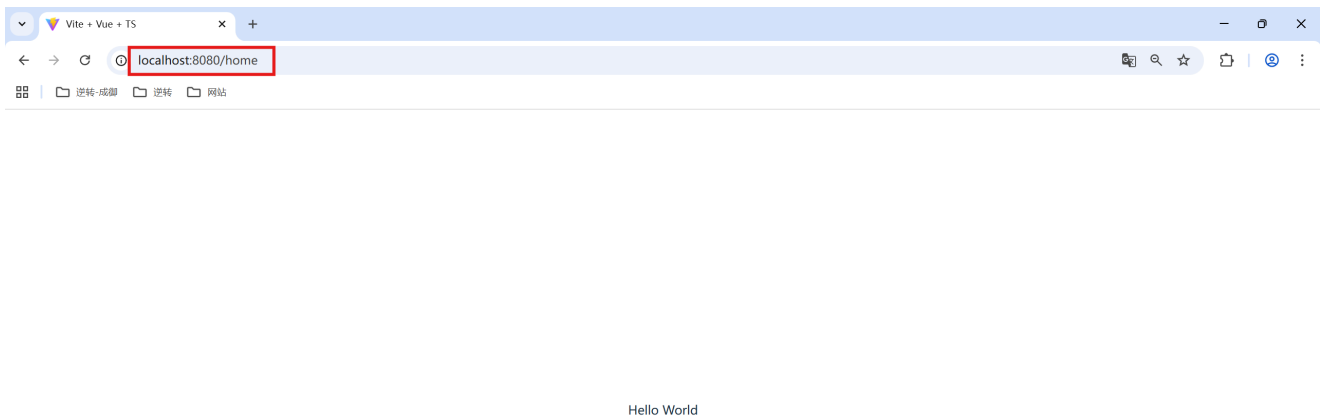
```
● PS E:\MyProgram\VUE\study_test2_example> pnpm install vite-tsconfig-paths
Packages: +5
+++++
Progress: resolved 103, reused 53, downloaded 4, added 5, done

dependencies:
+ vite-tsconfig-paths 5.1.4

Warning
Ignored build scripts: esbuild.
Run "pnpm approve-builds" to pick which dependencies should be allowed to run scripts.

Done in 1.3s using pnpm v10.14.0
```

最终页面显示如下：



3. 安装 Element Plus

3.1 安装

```
1  pnpm install element-plus --save
2  # pnpm install element-plus
3  pnpm install @element-plus/icons-vue
```

```
PS E:\MyProgram\VUE\study_test2_example> pnpm install element-plus --save
Packages: +22
```

```
+++++
Downloading element-plus@2.11.2: 7.55 MB/7.55 MB, done
Progress: resolved 125, reused 63, downloaded 16, added 22, done
```

```
dependencies:
+ element-plus 2.11.2
```

Warning

```
Ignored build scripts: vue-demi.
Run "pnpm approve-builds" to pick which dependencies should be allowed to run scripts.
```

```
Done in 17.9s using pnpm v10.14.0
```

```
PS E:\MyProgram\VUE\study_test2_example> pnpm install @element-plus/icons-vue
Already up to date
Progress: resolved 125, reused 79, downloaded 0, added 0, done
```

```
dependencies:
+ @element-plus/icons-vue 2.3.2
```

```
Done in 1.6s using pnpm v10.14.0
```

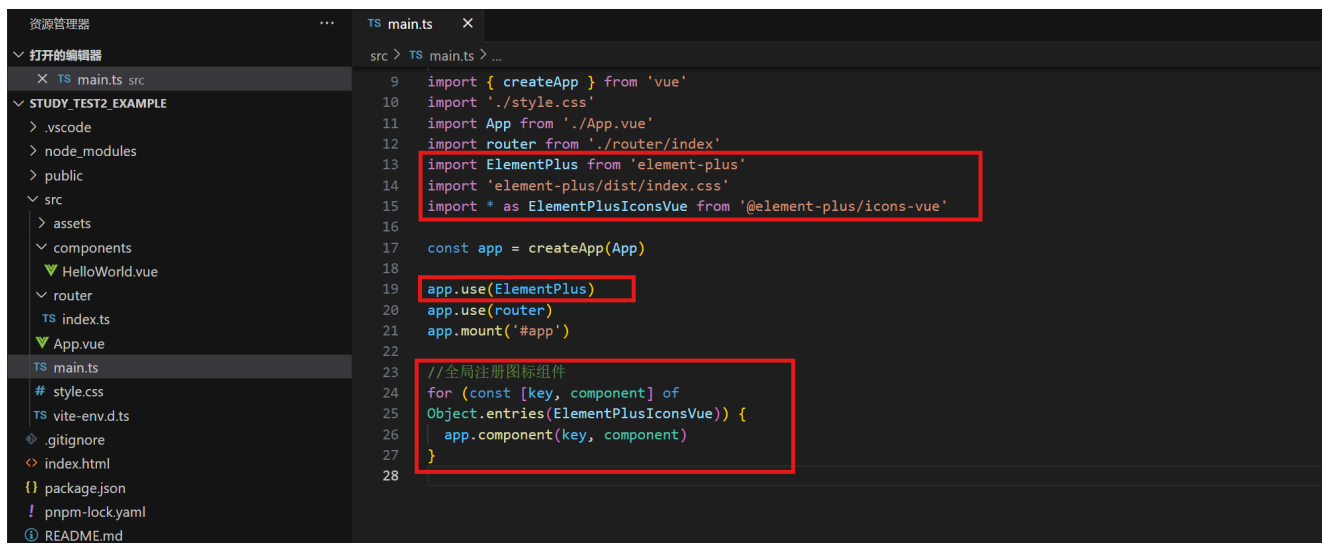
3.2 在 main.ts 中引入 Element Plus

```
1  import ElementPlus from 'element-plus'
```

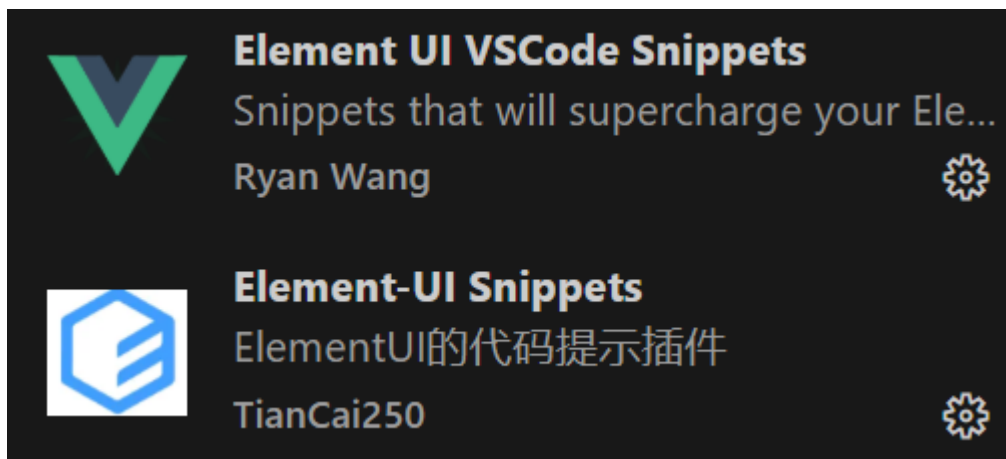
```

2  import 'element-plus/dist/index.css'
3  import * as ElementPlusIconsVue from '@element-plus/icons-vue'
4
5  app.use(ElementPlus)
6
7  //全局注册图标组件
8  for (const [key, component] of
9    Object.entries(ElementPlusIconsVue)) {
10    app.component(key, component)
11  }

```



3.3 安装相关插件



3.4 测试

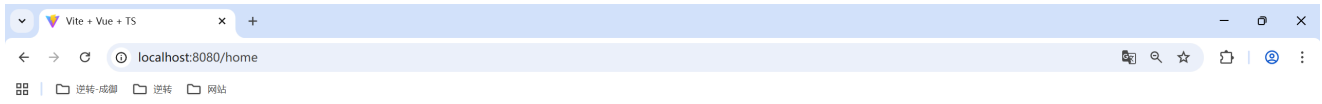
在 HelloWorld.vue 文件内修改代码为

```

1  <script setup lang="ts">
2
3  </script>
4
5  <template>
6    <div>Hello World</div>

```

```
7     <el-button type="primary">Primary Button</el-button>
8   </template>
9
10  <style lang="scss" scoped>
11
12  </style>
```



成功展示按钮，引入成功。

4. 初识 Pinia 与安装

官网：[Pinia | The intuitive store for Vue.js](https://pinia.vuejs.cn/)

4.1 安装依赖

```
1  pnpm add pinia
```

```
PS E:\MyProgram\Vue\study_test2_example> pnpm add pinia
Packages: +13
+++++
Progress: resolved 138, reused 82, downloaded 10, added 13, done

dependencies:
+ pinia 3.0.3

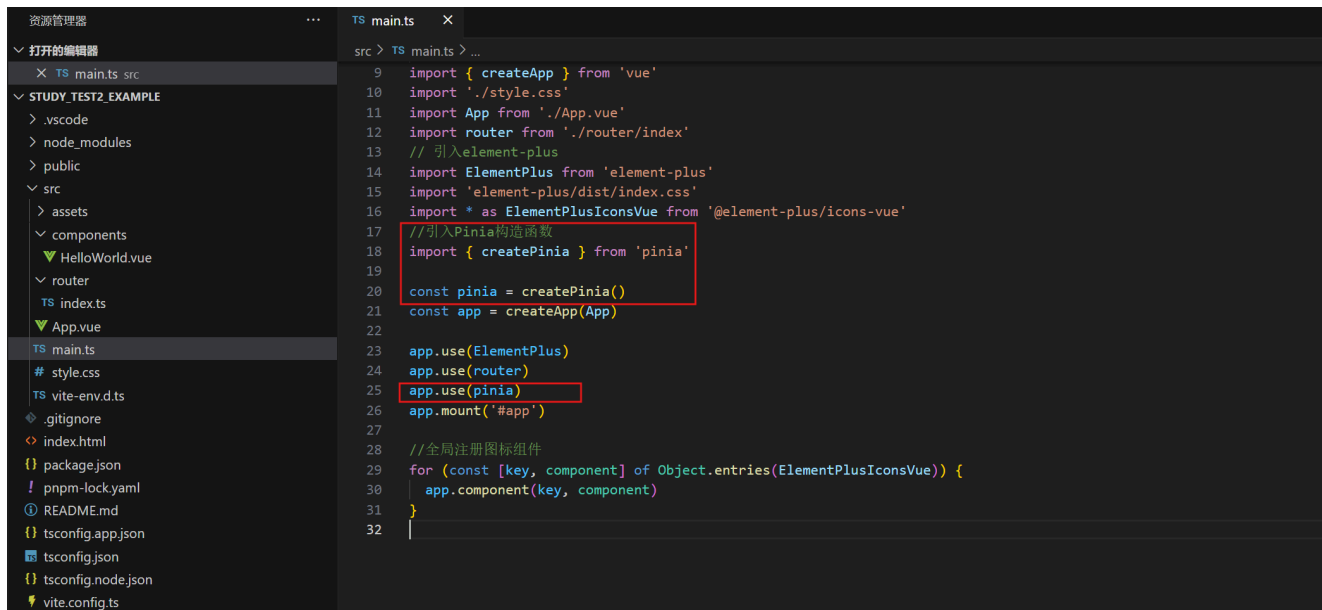
Done in 2.3s using pnpm v10.14.0
```

4.2 在 main.ts 引入 Pinia

```
1  //引入Pinia构造函数
2  import { createPinia } from 'pinia'
3
4  // 实例化 Pinia
5  const pinia = createPinia()
```

6

7 `app.use(pinia)`



4.3 使用 Pinia

4.3.1 定义仓库

在 `src` 下新建 `store` 文件夹，然后在 `store` 文件夹建立 `index.ts` 文件。

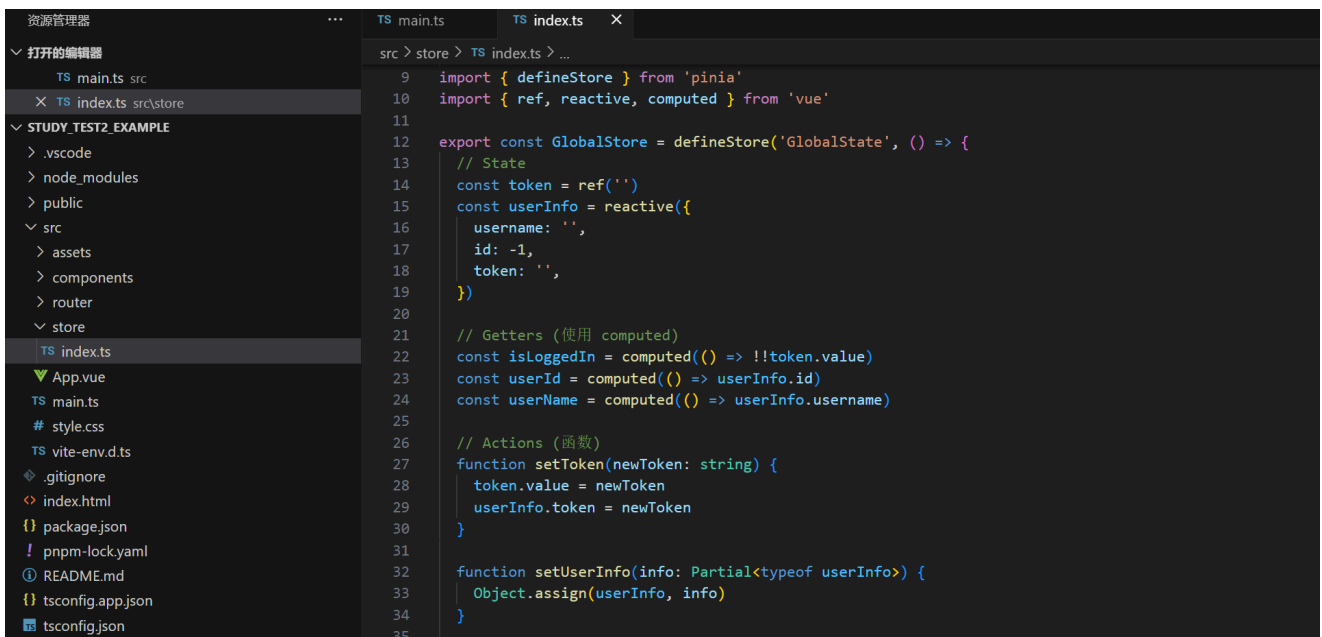
因为方才下载的 `pinia` 的版本是 3.x，故使用了 Getters 和 Actions 的新写法。若下载的 `pinia` 的版本是 2.x，则采用选项式或者组合式 API 的写法。

```
1 import { defineStore } from 'pinia'
2 import { ref, reactive, computed } from 'vue'
3
4 export const GlobalStore = defineStore('GlobalState', () => {
5   // State
6   const token = ref('')
7   const userInfo = reactive({
8     username: '',
9     id: -1,
10    token: '',
11  })
12
13  // Getters (使用 computed)
14  const isLoggedIn = computed(() => !!token.value)
15  const userId = computed(() => userInfo.id)
16  const userName = computed(() => userInfo.username)
17
18  // Actions (函数)
19  function setToken(newToken: string) {
20    token.value = newToken
21    userInfo.token = newToken
22  }
```

```

22     }
23
24     function setUserInfo(info: Partial<typeof userInfo>) {
25         Object.assign(userInfo, info)
26     }
27
28     function clearUser() {
29         token.value = ''
30         Object.assign(userInfo, {
31             username: '',
32             id: -1,
33             token: '',
34         })
35     }
36
37     return {
38         // State
39         token,
40         userInfo,
41
42         // Getters
43         isLoggedIn,
44         userId,
45         userName,
46
47         // Actions
48         setToken,
49         setUserInfo,
50         clearUser,
51     }
52 })

```

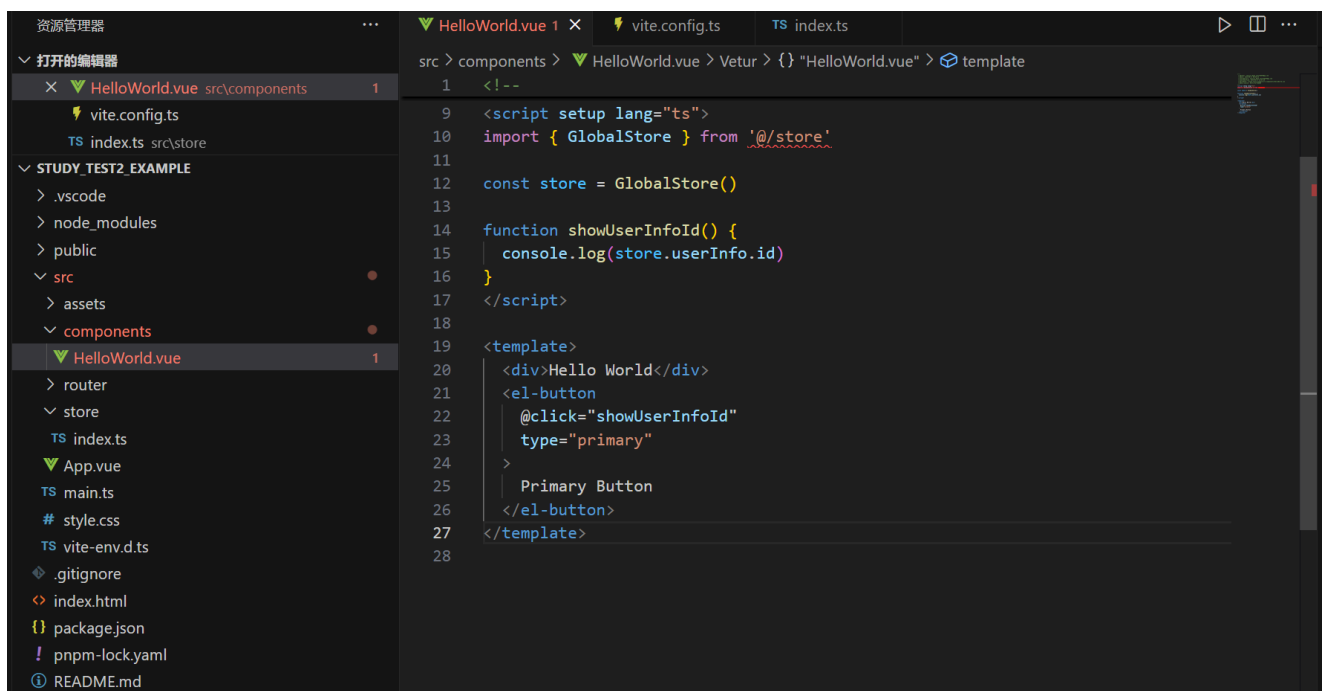


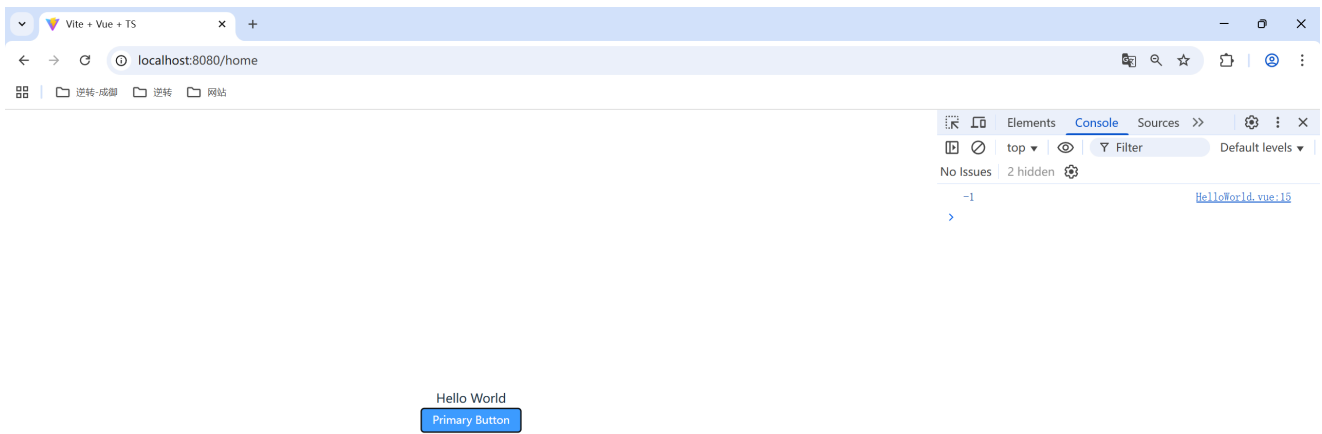
4.3.2 修改 HelloWorld.vue 并进行测试

```

1  <script setup lang="ts">
2  import { GlobalStore } from '@store'
3
4  const store = GlobalStore()
5
6  function showUserInfoId() {
7    console.log(store.userInfo.id)
8  }
9  </script>
10
11 <template>
12   <div>Hello World</div>
13   <el-button
14     @click="showUserInfoId"
15     type="primary"
16   >
17     Primary Button
18   </el-button>
19 </template>

```





5. 主界面布局

5.1 安装sass

```
1  pnpm add --save-dev sass
```

```
问题 输出 调试控制台 终端 端口 SPELL CHECKER
PS E:\MyProgram\VUE\study_test2_example> pnpm add --save-dev sass
Packages: +19 -3
+++++-----
Progress: resolved 166, reused 98, downloaded 10, added 19, done

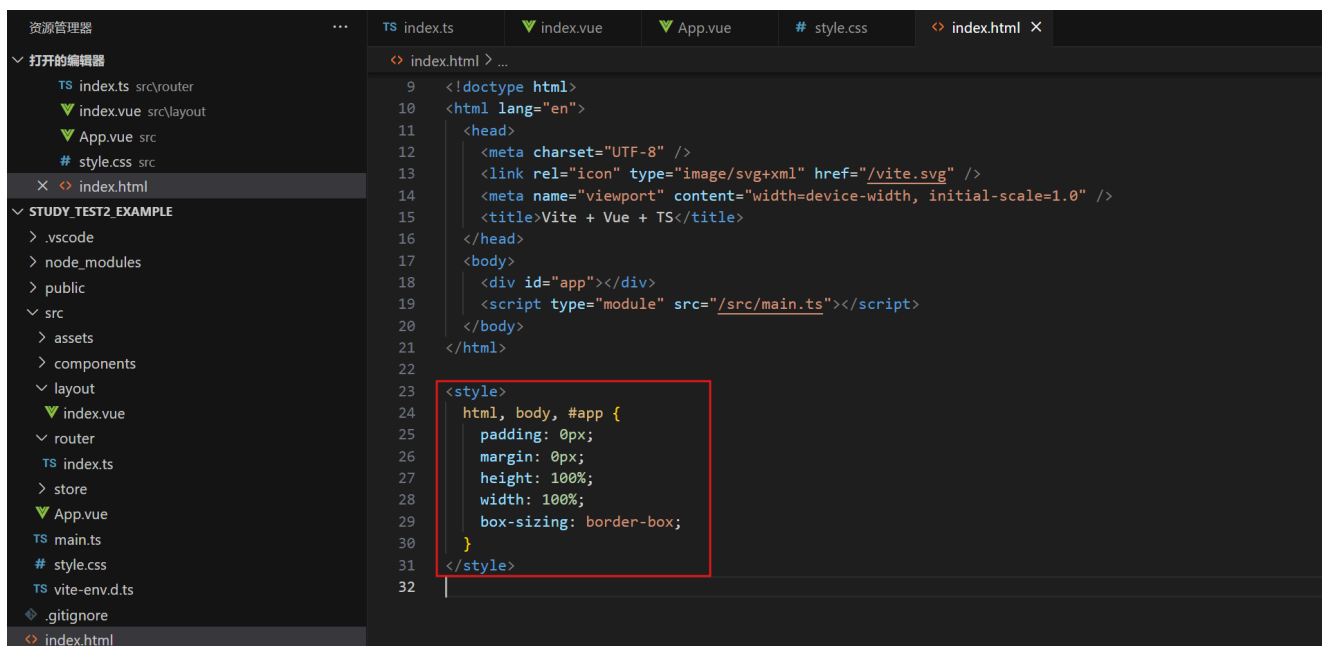
devDependencies:
+ sass 1.92.1

Warning
  Ignored build scripts: @parcel/watcher.
  Run "pnpm approve-builds" to pick which dependencies should be allowed to run scripts.

Done in 4.5s using pnpm v10.14.0
```

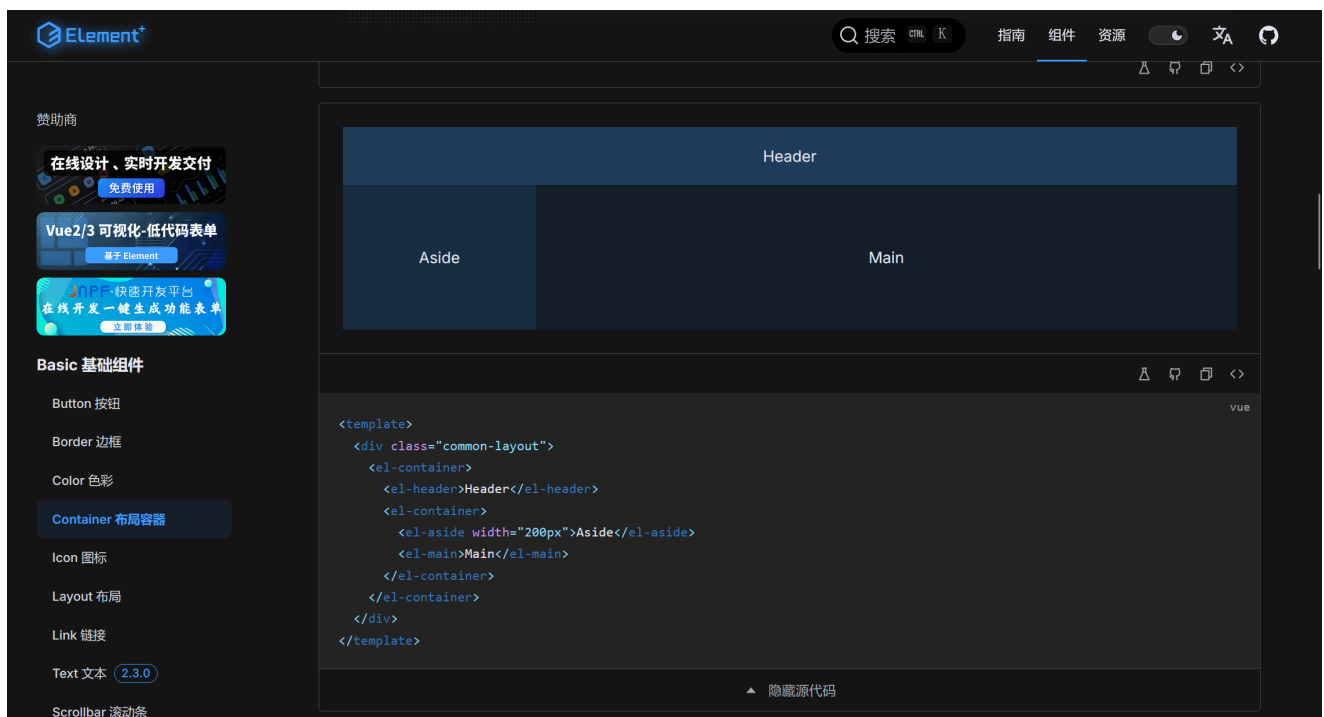
5.2 找到 index.html 添加如下 style 样式

```
1  <style>
2    html, body, #app {
3      padding: 0px;
4      margin: 0px;
5      height: 100%;
6      width: 100%;
7      box-sizing: border-box;
8    }
9  </style>
```



5.3 在 src 目录下新建 layout 目录，并新建 index.vue 主页面组件

从 Element Plus 官网（<https://element-plus.sxtxhy.com/zh-CN/>）复制页面布局。

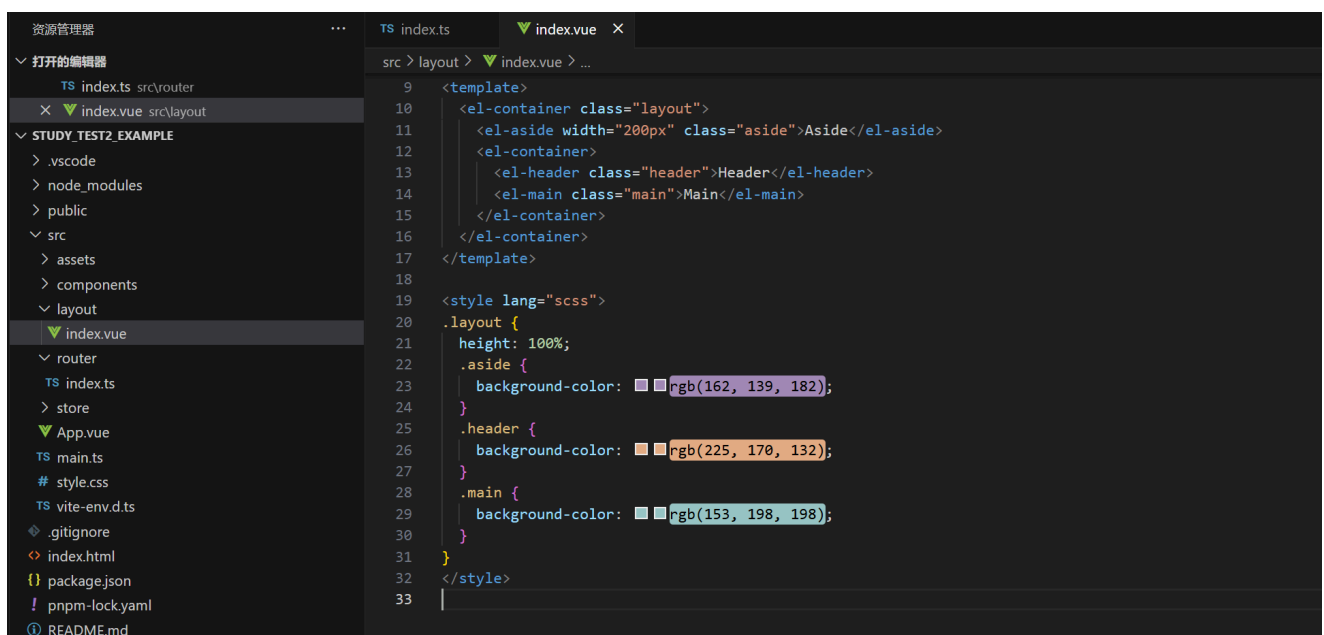


```
1 <template>
2   <el-container class="layout">
3     <el-aside width="200px" class="aside">Aside</el-aside>
4     <el-container>
5       <el-header class="header">Header</el-header>
6       <el-main class="main">Main</el-main>
7     </el-container>
8   </el-container>
9 </template>
10
```

```

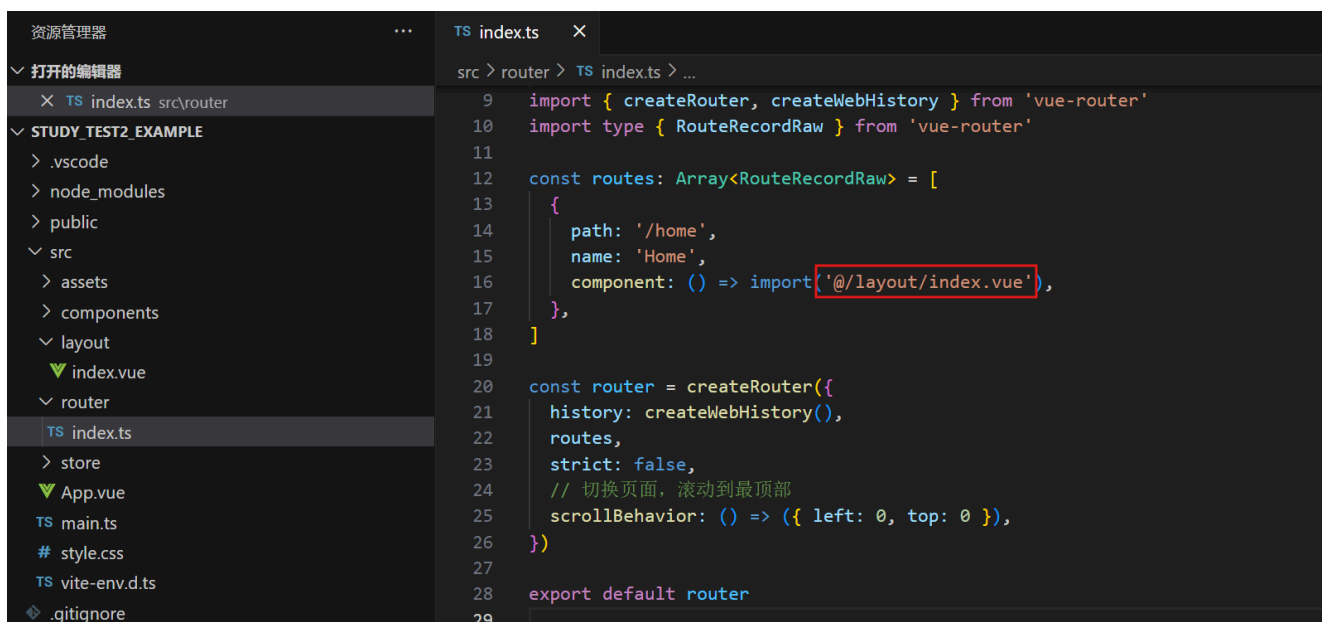
11 <style lang="scss">
12 .layout {
13     height: 100%;
14     .aside {
15         background-color: rgb(162, 139, 182);
16     }
17     .header {
18         background-color: rgb(225, 170, 132);
19     }
20     .main {
21         background-color: rgb(153, 198, 198);
22     }
23 }
24 </style>

```

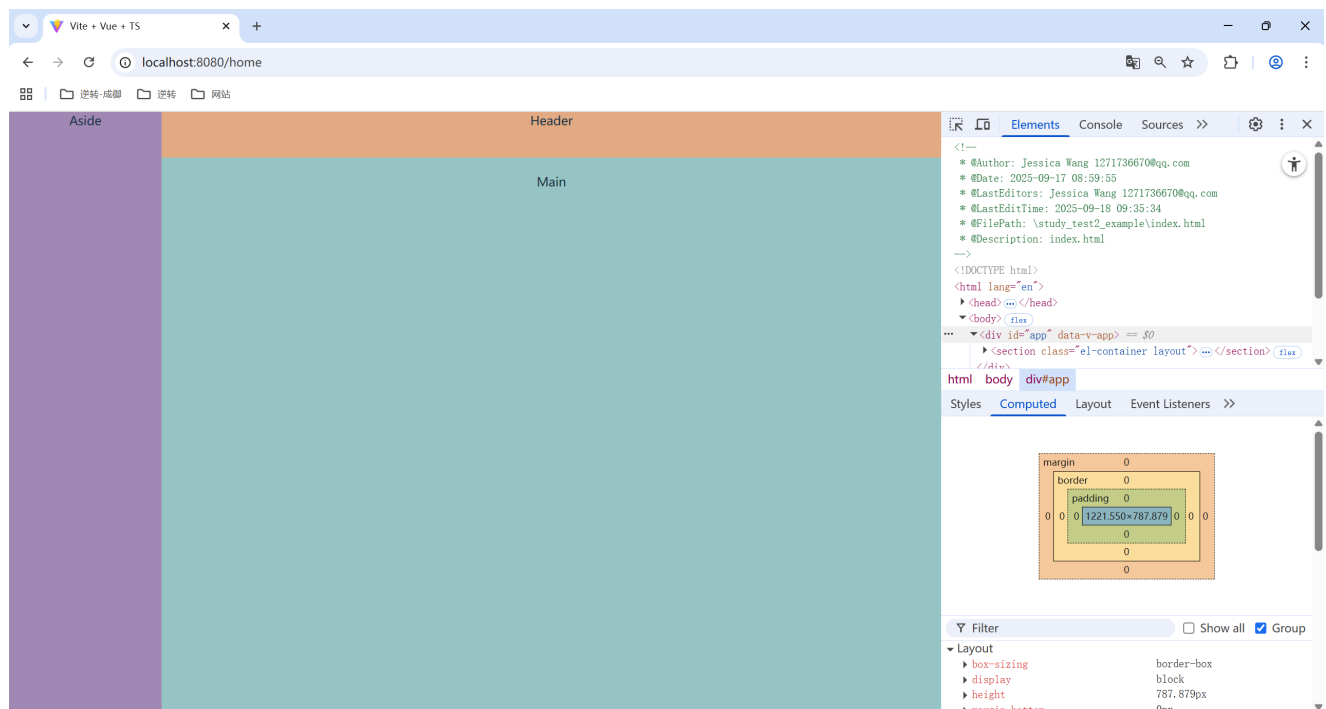


5.4 在 router 中引入主页面组件

在 router 目录下的 index.ts 中修改路径。



启动项目，查看效果。



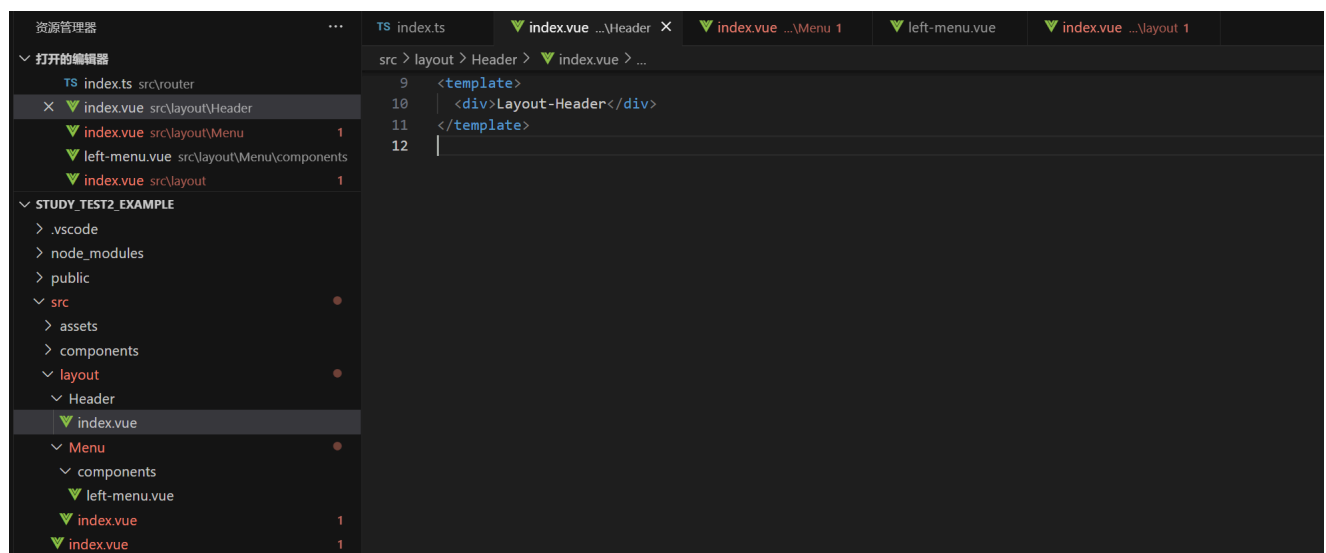
6 左侧导航菜单制作

对应的 Element Plus 组件库中的菜单：[Menu 菜单 | Element Plus](#)

6.1 创建 Header 组件

在 layout 文件夹下新建 Header 目录，然后新建 `index.vue` 组件

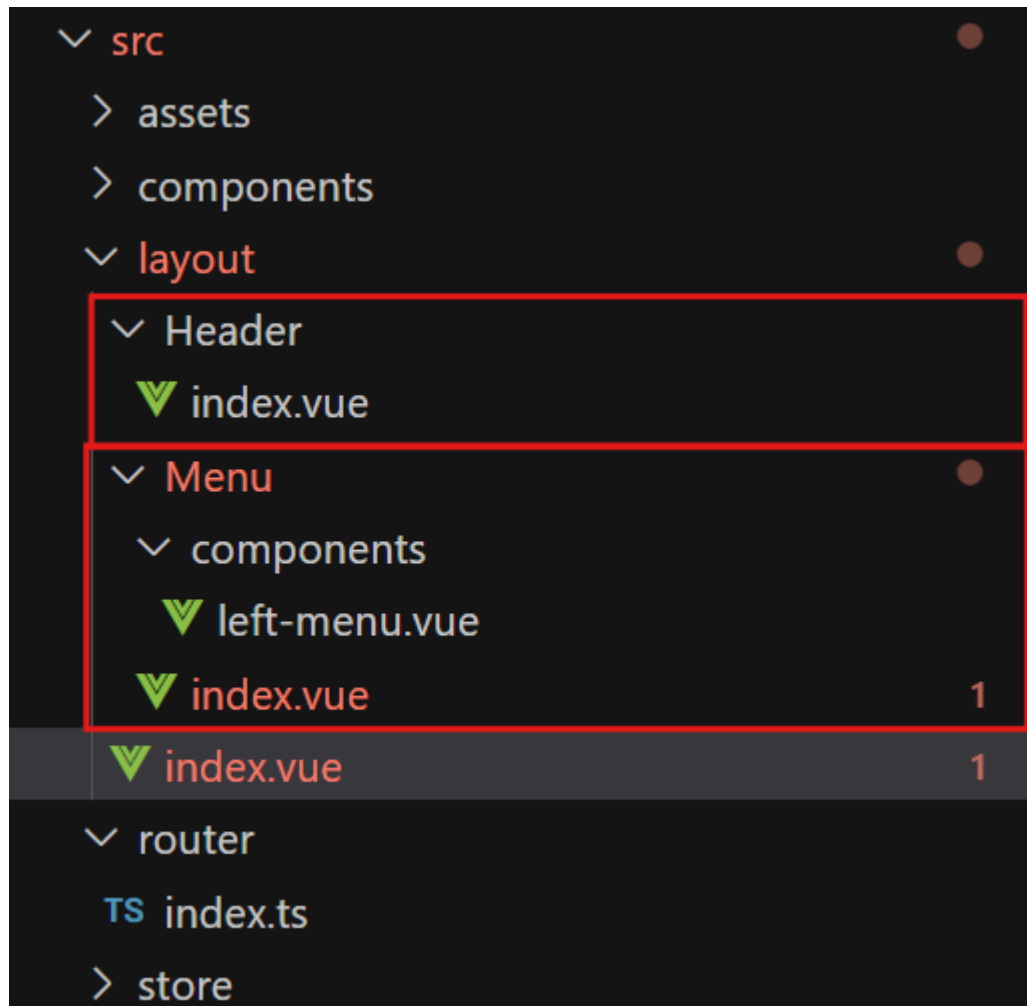
```
1 <template>
2   <div>Layout-Header</div>
3 </template>
```



6.2 创建 Menu 目录

在 layout 文件夹下新建 Menu 目录，然后新建 index.vue 组件和 components 文件夹中的 left-menu.vue 组件。

目录结构如下。



6.2.1 layout\Menu\index.vue 文件

```
1  <script setup lang="ts">
2  import { ref, reactive } from "vue"
3  import LeftMenu from "../components/left-menu.vue"
4
5  const isCollapse = ref(false);
6  const menuList = reactive([
7    {
8      path: "/system",
9      component: "Layout",
10     name: "system",
11     meta: {
12       title: "系统管理",
13       icon: "Setting",
14       roles: ["sys:manage"],
15     },
16     children: [
17       {
```

```
18     path: "/userList",
19     component: "/system/User/UserList",
20     name: "userList",
21     meta: {
22         title: "员工管理",
23         icon: "UserFilled",
24         roles: ["sys:user"],
25     },
26 },
27 {
28     path: "/roleList",
29     component: "/system/Role/RoleList",
30     name: "roleList",
31     meta: {
32         title: "角色管理",
33         icon: "Wallet",
34         roles: ["sys:role"],
35     },
36 },
37 {
38     path: "/menuList",
39     component: "/system/Menu/MenuList",
40     name: "menuList",
41     meta: {
42         title: "菜单管理",
43         icon: "Menu",
44         roles: ["sys:menu"],
45     },
46 },
47 ],
48 },
49 {
50     path: "/memberRoot",
51     component: "Layout",
52     name: "memberRoot",
53     meta: {
54         title: "会员管理",
55         icon: "Setting",
56         roles: ["sys:memberRoot"],
57     },
58     children: [
59         {
60             path: "/cardType",
61             component: "/member/CardType",
62             name: "cardType",
63             meta: {
64                 title: "会员卡类型",
65                 icon: "UserFilled",
66                 roles: ["sys:cardType"],
```

```

67         },
68     },
69     {
70         path: "/memberList",
71         component: "/member/MemberList",
72         name: "memberList",
73         meta: {
74             title: "会员管理",
75             icon: "Wallet",
76             roles: ["sys:memberList"],
77         },
78     },
79     {
80         path: "/myFee",
81         component: "/system/FeeList",
82         name: "myFee",
83         meta: {
84             title: "我的充值",
85             icon: "Menu",
86             roles: ["sys:myFee"],
87         },
88     },
89 ],
90 },
91 ])
92 </script>
93
94 <template>
95     <el-menu
96         default-active="2"
97         class="el-menu-vertical-demo"
98         :collapse="isCollapse"
99         unique-opened
100         background-color="#304156"
101     >
102         <left-menu :menuList="menuList" />
103     </el-menu>
104 </template>
105
106 <style lang="scss" scoped>
107 .el-menu-vertical-demo:not(.el-menu--collapse) {
108     width: 230px;
109     min-height: 400px;
110 }
111 .el-menu {
112     height: 100%;
113     border-right: none;
114 }
115 :deep(.el-sub-menu .el-sub-menu__title){

```

```

116     color: #f4f4f5 !important;
117 }
118 :deep(.el-menu .el-menu-item){
119     color: #bfcabd9;
120 }
121 /* 菜单点中文字的颜色 */
122 :deep(.el-menu-item.is-active){
123     color: #409eff !important;
124 }
125 /* 当前打开菜单的所有子菜单颜色 */
126 :deep(.is-opened .el-menu-item){
127     background-color: #1f2d3d !important;
128 }
129 /* 鼠标移动菜单的颜色 */
130 :deep(.el-menu-item:hover){
131     background-color: #001528 !important;
132 }
133 </style>

```

6.2.2 layout\Menu\components\left-menu.vue 文件

```

1  <script setup lang="ts">
2  defineProps(["menuList"]);
3  </script>
4
5  <template>
6      <template
7          v-for="menu in menuList"
8          :key="menu.path"
9      >
10         <el-sub-menu
11             v-if="menu.children && menu.children.length > 0"
12             :index="menu.path"
13         >
14             <template #title>
15                 <el-icon>
16                     <component :is="menu.meta.icon"></component>
17                 </el-icon>
18                 <span>{{ menu.meta.title }}</span>
19             </template>
20             <left-menu :menuList="menu.children"></left-menu>
21         </el-sub-menu>
22         <el-menu-item
23             v-else
24             style="color: #f4f4f5"
25             :index="menu.path"
26         >
27             <el-icon>

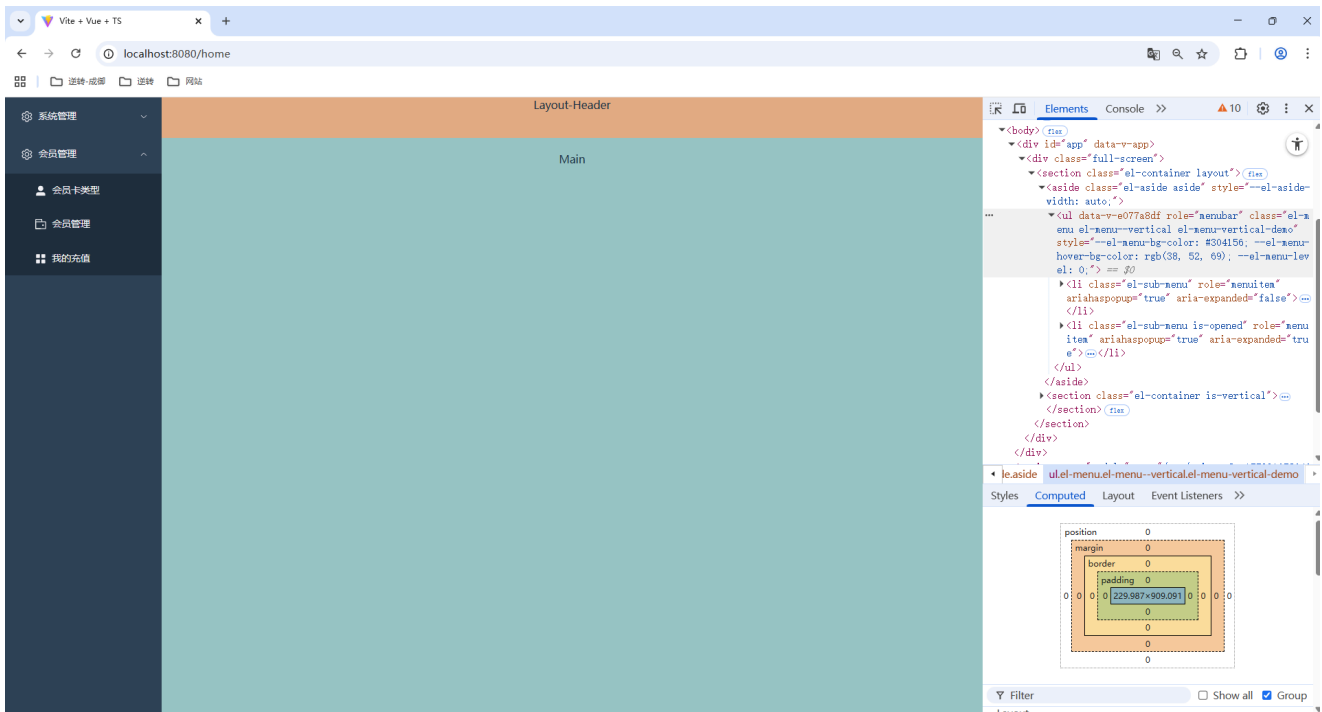
```

```
28         <component :is="menu.meta.icon"></component>
29     </el-icon>
30     <template #title>{{ menu.meta.title }}</template>
31 </el-menu-item>
32 </template>
33 </template>
```

6.3 修改 layout/index.vue 组件

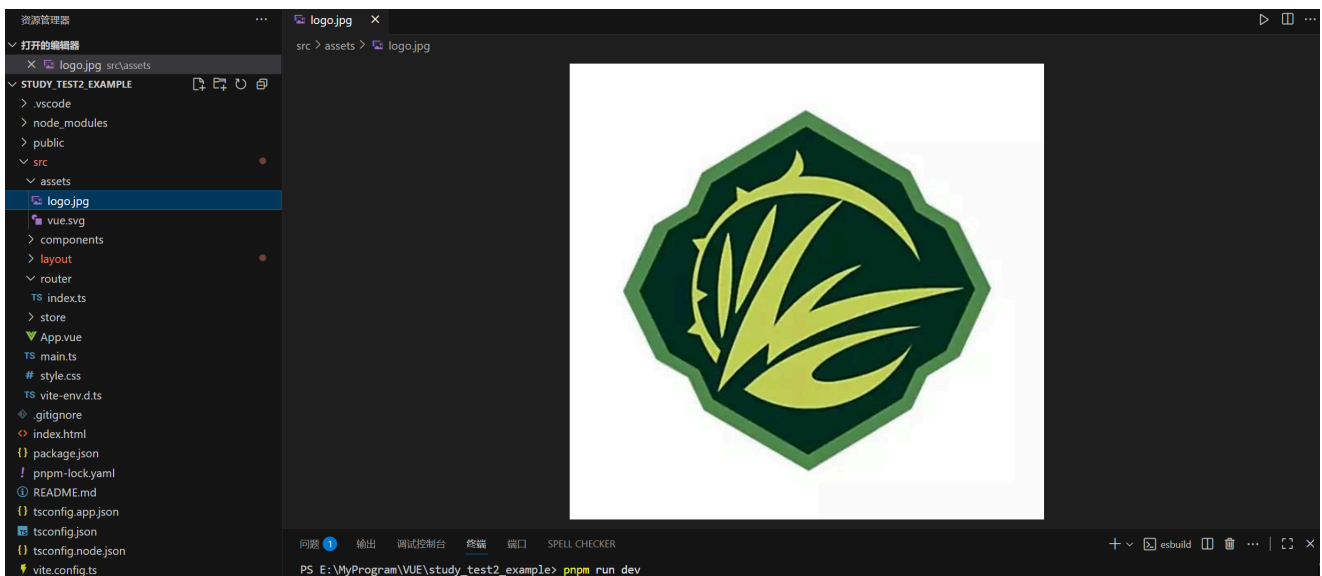
```
1  <script lang="ts" setup>
2  import CustomHeader from './Header/index.vue'
3  import CustomMenu from './Menu/index.vue'
4  </script>
5
6  <template>
7      <div class="full-screen">
8          <el-container class="layout">
9              <el-aside width="auto" class="aside">
10                 <custom-menu />
11             </el-aside>
12             <el-container>
13                 <el-header class="header">
14                     <custom-header />
15                 </el-header>
16                 <el-main class="main">Main</el-main>
17             </el-container>
18         </el-container>
19     </div>
20 </template>
21
22 <style lang="scss">
23 .full-screen {
24     height: 100vh; /* 视口高度 */
25     width: 100vw; /* 视口宽度 */
26 }
27 .layout {
28     height: 100%;
29     .aside {
30         height: 100%;
31         background-color: rgb(162, 139, 182);
32     }
33     .header {
34         background-color: rgb(225, 170, 132);
35     }
36     .main {
37         background-color: rgb(153, 198, 198);
38     }
39 }
```

6.4 成果展示



7.菜单logo制作

7.1 项目 assets 里面加入logo



7.2 修改 layout\Menu\index.vue 文件

添加 logo 和项目名称。

```
1 <script setup lang="ts">
2 import { ref, reactive } from "vue"
3 import LeftMenu from "../components/left-menu.vue"
4 import MenuLogo from '@assets/logo.jpg'
```

```
5
6 const isCollapse = ref(false);
7 const menuList = reactive([
8   {
9     path: "/system",
10    component: "Layout",
11    name: "system",
12    meta: {
13      title: "系统管理",
14      icon: "Setting",
15      roles: ["sys:manage"],
16    },
17    children: [
18      {
19        path: "/userList",
20        component: "/system/User/UserList",
21        name: "userList",
22        meta: {
23          title: "员工管理",
24          icon: "UserFilled",
25          roles: ["sys:user"],
26        },
27      },
28      {
29        path: "/roleList",
30        component: "/system/Role/RoleList",
31        name: "roleList",
32        meta: {
33          title: "角色管理",
34          icon: "Wallet",
35          roles: ["sys:role"],
36        },
37      },
38      {
39        path: "/menuList",
40        component: "/system/Menu/MenuList",
41        name: "menuList",
42        meta: {
43          title: "菜单管理",
44          icon: "Menu",
45          roles: ["sys:menu"],
46        },
47      },
48    ],
49  },
50  {
51    path: "/memberRoot",
52    component: "Layout",
53    name: "memberRoot",
```

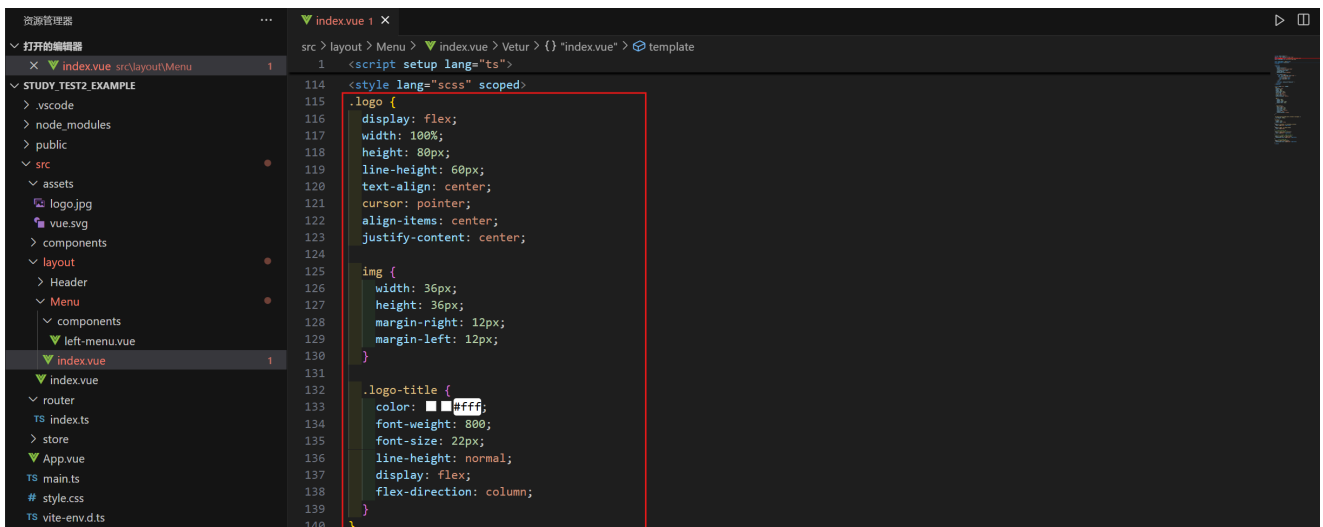
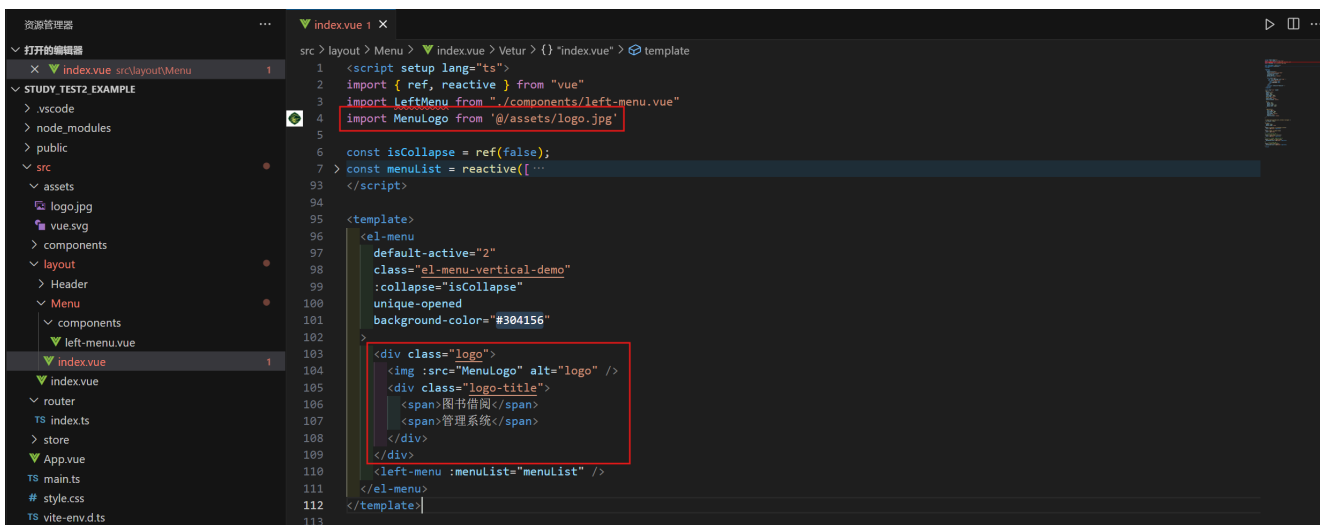
```
54     meta: {
55         title: "会员管理",
56         icon: "Setting",
57         roles: ["sys:memberRoot"],
58     },
59     children: [
60         {
61             path: "/cardType",
62             component: "/member/CardType",
63             name: "cardType",
64             meta: {
65                 title: "会员卡类型",
66                 icon: "UserFilled",
67                 roles: ["sys:cardType"],
68             },
69         },
70         {
71             path: "/memberList",
72             component: "/member/MemberList",
73             name: "memberList",
74             meta: {
75                 title: "会员管理",
76                 icon: "Wallet",
77                 roles: ["sys:memberList"],
78             },
79         },
80         {
81             path: "/myFee",
82             component: "/system/FeeList",
83             name: "myFee",
84             meta: {
85                 title: "我的充值",
86                 icon: "Menu",
87                 roles: ["sys:myFee"],
88             },
89         },
90     ],
91 },
92 ])
93 </script>
94
95 <template>
96     <el-menu
97         default-active="2"
98         class="el-menu-vertical-demo"
99         :collapse="isCollapse"
100         unique-opened
101         background-color="#304156"
102     >
```

```
103     <div class="logo">
104         
105         <div class="logo-title">
106             <span>图书借阅</span>
107             <span>管理系统</span>
108         </div>
109     </div>
110     <left-menu :menuList="menuList" />
111 </el-menu>
112 </template>
113
114 <style lang="scss" scoped>
115 .logo {
116     display: flex;
117     width: 100%;
118     height: 80px;
119     line-height: 60px;
120     text-align: center;
121     cursor: pointer;
122     align-items: center;
123     justify-content: center;
124
125     img {
126         width: 36px;
127         height: 36px;
128         margin-right: 12px;
129         margin-left: 12px;
130     }
131
132     .logo-title {
133         color: #fff;
134         font-weight: 800;
135         font-size: 22px;
136         line-height: normal;
137         display: flex;
138         flex-direction: column;
139     }
140 }
141
142 .el-menu-vertical-demo:not(.el-menu--collapse) {
143     min-height: 400px;
144 }
145 .el-menu {
146     height: 100%;
147     border-right: none;
148 }
149 :deep(.el-sub-menu .el-sub-menu__title){
150     color: #f4f4f5 !important;
151 }
```

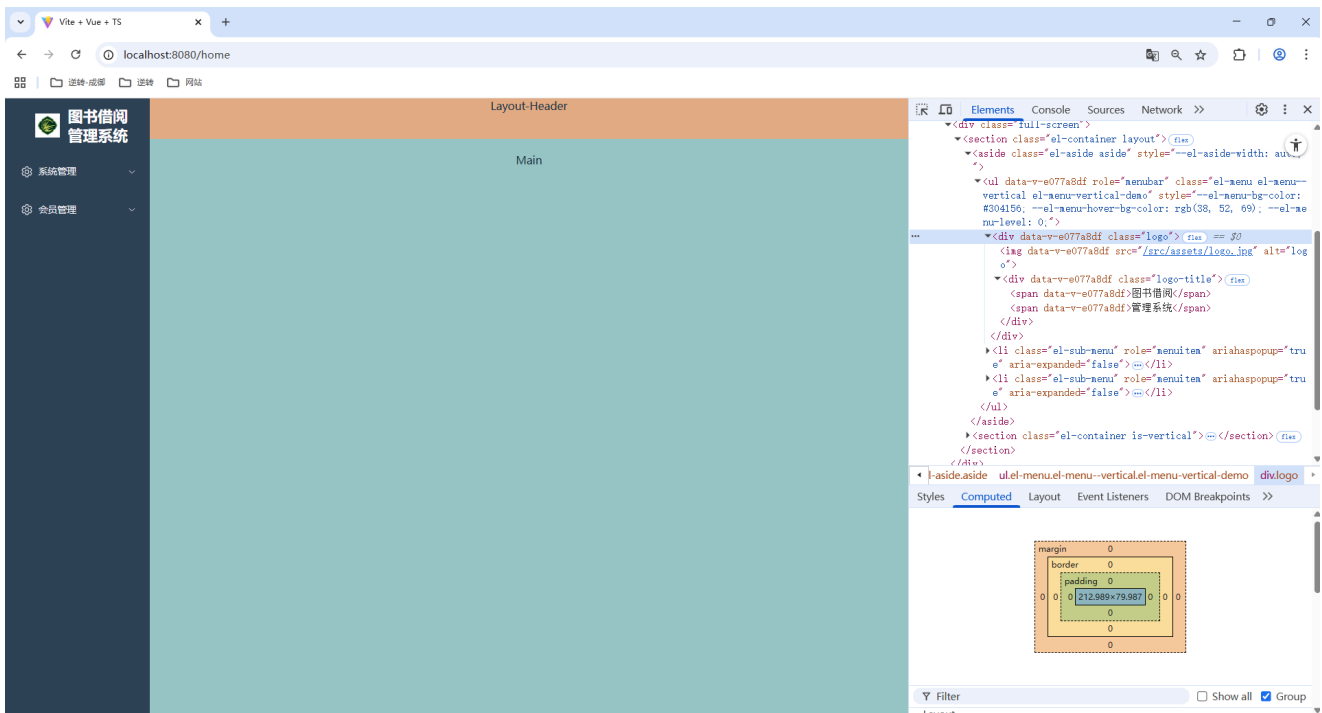
```

152 :deep(.el-menu .el-menu-item){
153   color: #bfcabd9;
154 }
155 /* 菜单点中文字的颜色 */
156 :deep(.el-menu-item.is-active){
157   color: #409eff !important;
158 }
159 /* 当前打开菜单的所有子菜单颜色 */
160 :deep(.is-opened .el-menu-item){
161   background-color: #1f2d3d !important;
162 }
163 /* 鼠标移动菜单的颜色 */
164 :deep(.el-menu-item:hover){
165   background-color: #001528 !important;
166 }
167 </style>

```



7.3 成果展示



8.路由配置与页面创建

要求：点击左侧菜单，能够在内容展示区展示对应页面。

8.1 修改路由

router/index.ts 路由修改为如下。

```
1  import { createRouter, createWebHistory } from 'vue-router'
2  import type { RouteRecordRaw } from 'vue-router'
3
4  const routes: Array<RouteRecordRaw> = [
5    {
6      path: '/',
7      redirect: '/home',
8    },
9    {
10     path: '/home',
11     component: () => import('@/layout/index.vue'),
12     redirect: '/home',
13     children: [
14       {
15         path: '/home',
16         component: () => import('@/views/home/index.vue'),
17         name: 'home',
18         meta: {
19           title: '首页',
20           icon: '#iconhome',
21         },
22       },
23     ],
24   },
25 ]
```

```
24     },
25     {
26       path: "/system",
27       component: () => import('@/layout/index.vue'),
28       name: "system",
29       meta: {
30         title: "系统管理",
31         icon: "el-icon-menu",
32         roles: ["sys:manage"]
33       },
34       children: [
35         {
36           path: "/userList",
37           component: () => import('@/views/system/UserList.vue'),
38           name: "userList",
39           meta: {
40             title: "员工管理",
41             icon: "el-icon-s-custom",
42             roles: ["sys:user"]
43           },
44         },
45         {
46           path: "/roleList",
47           component: () => import('@/views/system/RoleList.vue'),
48           name: "roleList",
49           meta: {
50             title: "角色管理",
51             icon: "el-icon-s-tools",
52             roles: ["sys:role"]
53           },
54         },
55         {
56           path: "/menuList",
57           component: () => import('@/views/system/MenuList.vue'),
58           name: "menuList",
59           meta: {
60             title: "权限管理",
61             icon: "el-icon-document",
62             roles: ["sys:menu"]
63           },
64         },
65       ]
66     },
67     {
68       path: "/memberRoot",
69       component: () => import('@/layout/index.vue'),
70       name: "memberRoot",
71       meta: {
72         title: "会员管理",
```

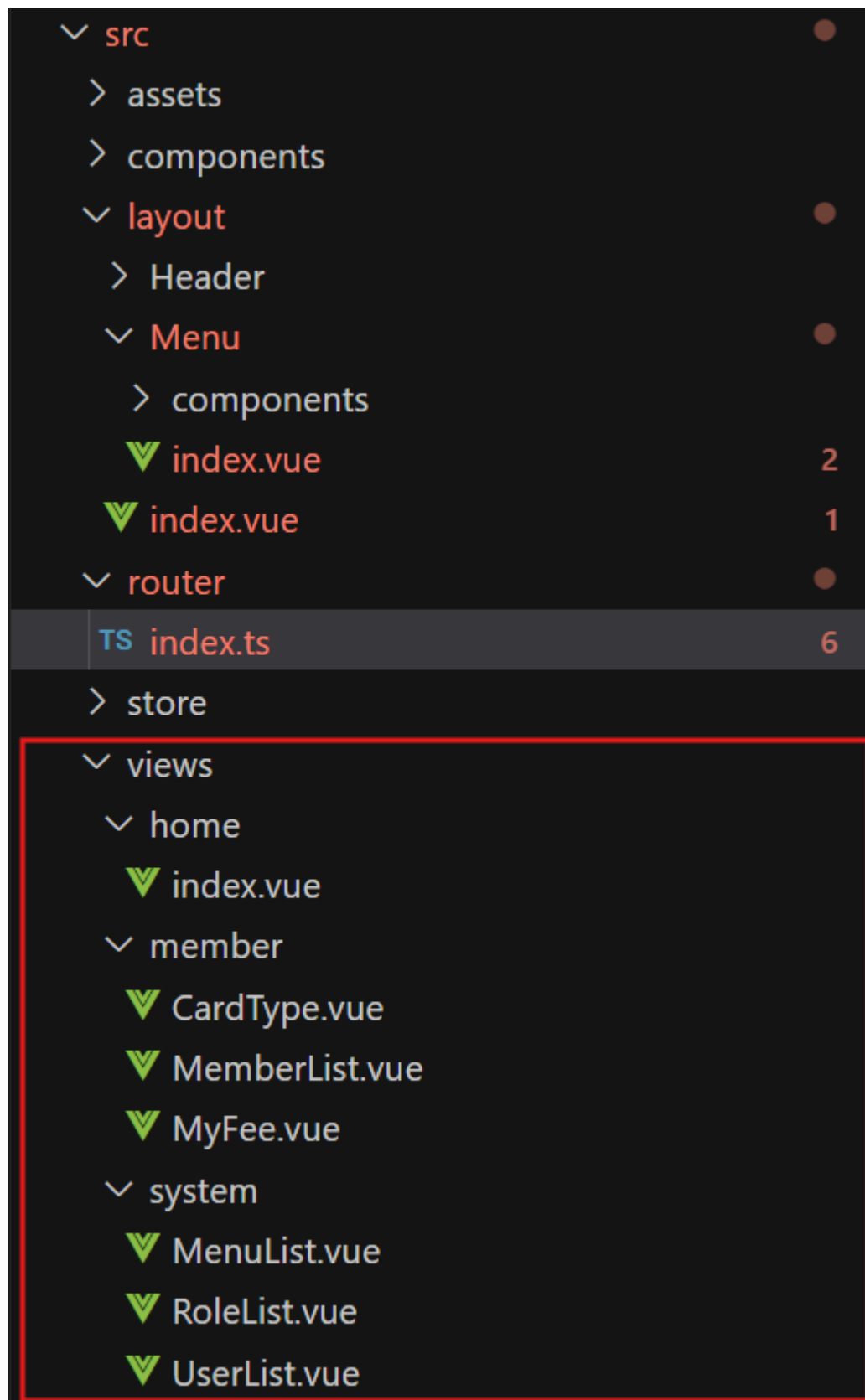
```

73     icon: "Setting",
74     roles: ["sys:memberRoot"],
75 },
76 children: [
77   {
78     path: "/cardType",
79     component: () => import('@/views/member/CardType.vue'),
80     name: "cardType",
81     meta: {
82       title: "会员卡类型",
83       icon: "UserFilled",
84       roles: ["sys:cardType"],
85     },
86   },
87   {
88     path: "/memberList",
89     component: () => import('@/views/member/MemberList.vue'),
90     name: "memberList",
91     meta: {
92       title: "会员管理",
93       icon: "Wallet",
94       roles: ["sys:memberList"],
95     },
96   },
97   {
98     path: "/myFee",
99     component: () => import('@/views/member/MyFee.vue'),
100    name: "myFee",
101    meta: {
102      title: "我的充值",
103      icon: "Menu",
104      roles: ["sys:myFee"],
105    },
106  },
107 ],
108 }
109 ]
110
111 const router = createRouter({
112   history: createWebHistory(),
113   routes,
114   strict: false,
115   // 切换页面，滚动到最顶部
116   scrollBehavior: () => ({ left: 0, top: 0 }),
117 })
118
119 export default router

```

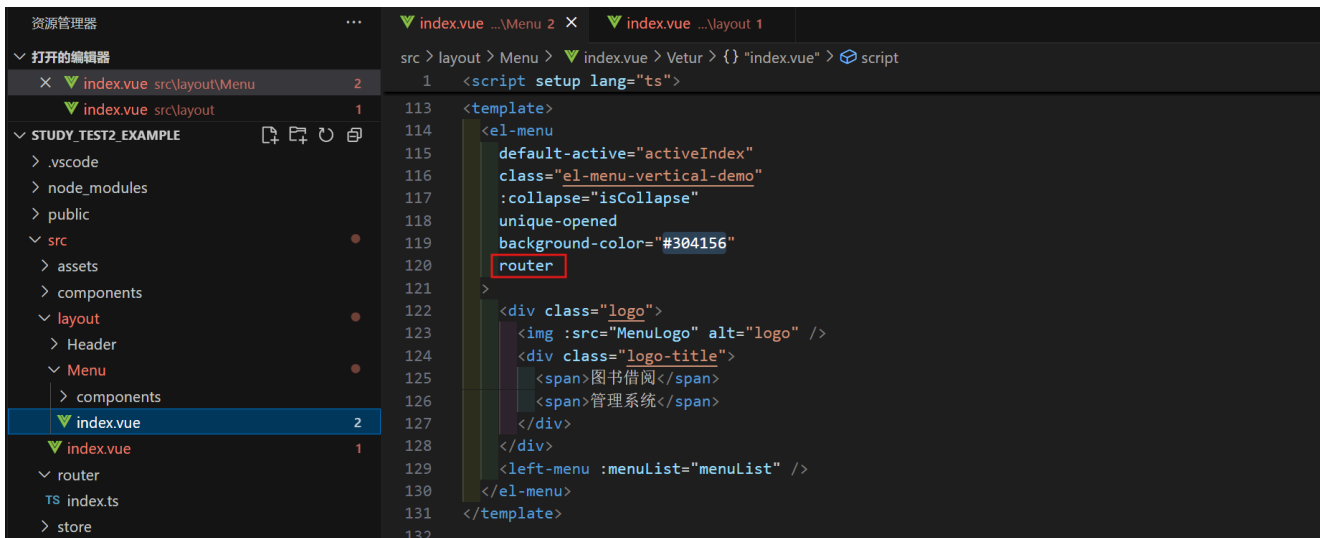
8.2 创建路由对应的页面

新建 `views` 文件夹，然后创建对应 `vue` 文件。

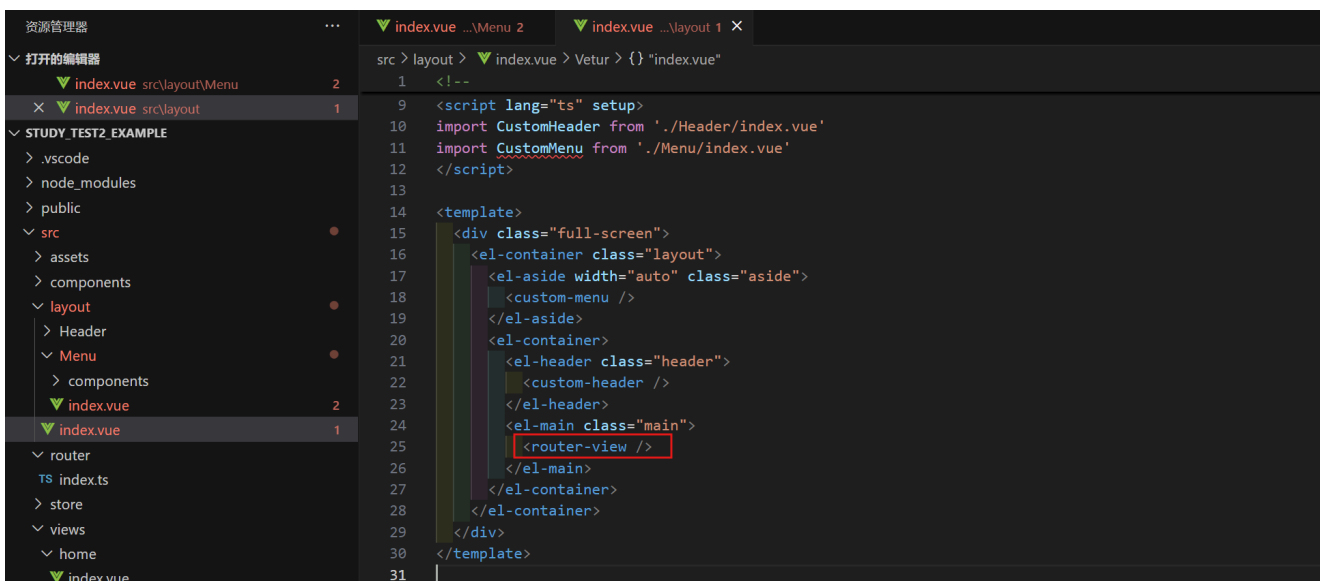


8.3 在 `layout\Menu\index.vue` 组件的 `<el-menu>` 添加 `router` 属性

`router`: 是否启用 `vue-router` 模式。启用该模式会在激活导航时以 `index` 作为 `path` 进行路由跳转。



8.4 在 layout\index.vue 的添加路由



```
1 <script lang="ts" setup>
2   import CustomHeader from './Header/index.vue'
3   import CustomMenu from './Menu/index.vue'
4 </script>
5
6 <template>
7   <div class="full-screen">
8     <el-container class="layout">
9       <el-aside width="auto" class="aside">
10         <custom-menu />
11       </el-aside>
12       <el-container>
13         <el-header class="header">
14           <custom-header />
15         </el-header>
16         <el-main class="main">
17           <router-view />
18         </el-main>
19       </el-container>
```

```

20     </el-container>
21   </div>
22 </template>
23
24 <style lang="scss">
25 .full-screen {
26   height: 100vh; /* 视口高度 */
27   width: 100vw; /* 视口宽度 */
28 }
29 .layout {
30   height: 100%;
31   .aside {
32     height: 100%;
33     width: 213px;
34     background-color: rgb(162, 139, 182);
35   }
36   .header {
37     background-color: rgb(225, 170, 132);
38   }
39   .main {
40     background-color: rgb(153, 198, 198);
41   }
42 }
43 </style>

```

8.5 在 layout\Menu\index.vue 组件设置当前激活的菜单

```

1  <script setup lang="ts">
2  import { ref, reactive, computed } from "vue"
3  import { useRoute } from "vue-router"
4  import LeftMenu from "../components/left-menu.vue"
5  import MenuLogo from '@assets/logo.jpg'
6
7  const route = useRoute()
8  const isCollapse = ref(false)
9  const menuList = reactive([
10    {
11      path: "/home",
12      component: "Layout",
13      name: "home",
14      meta: {
15        title: "首页",
16        icon: "House",
17        roles: ["sys:home"],
18      },
19    },
20    {
21      path: "/system",

```

```
22     component: "Layout",
23     name: "system",
24     meta: {
25         title: "系统管理",
26         icon: "Setting",
27         roles: ["sys:manage"],
28     },
29     children: [
30         {
31             path: "/userList",
32             component: "/system/User/UserList",
33             name: "userList",
34             meta: {
35                 title: "员工管理",
36                 icon: "UserFilled",
37                 roles: ["sys:user"],
38             },
39         },
40         {
41             path: "/roleList",
42             component: "/system/Role/RoleList",
43             name: "roleList",
44             meta: {
45                 title: "角色管理",
46                 icon: "Wallet",
47                 roles: ["sys:role"],
48             },
49         },
50         {
51             path: "/menuList",
52             component: "/system/Menu/MenuList",
53             name: "menuList",
54             meta: {
55                 title: "菜单管理",
56                 icon: "Menu",
57                 roles: ["sys:menu"],
58             },
59         },
60     ],
61 },
62 {
63     path: "/memberRoot",
64     component: "Layout",
65     name: "memberRoot",
66     meta: {
67         title: "会员管理",
68         icon: "Setting",
69         roles: ["sys:memberRoot"],
70     },
```

```

71     children: [
72       {
73         path: "/cardType",
74         component: "/member/CardType",
75         name: "cardType",
76         meta: {
77           title: "会员卡类型",
78           icon: "UserFilled",
79           roles: ["sys:cardType"],
80         },
81       },
82       {
83         path: "/memberList",
84         component: "/member/MemberList",
85         name: "memberList",
86         meta: {
87           title: "会员管理",
88           icon: "Wallet",
89           roles: ["sys:memberList"],
90         },
91       },
92       {
93         path: "/myFee",
94         component: "/system/FeeList",
95         name: "myFee",
96         meta: {
97           title: "我的充值",
98           icon: "Menu",
99           roles: ["sys:myFee"],
100        },
101      },
102    ],
103  },
104 ])
105
106 //获取激活的菜单
107 const activeIndex = computed(()=>{
108   const {path} = route;
109   return path;
110 })
111 </script>
112
113 <template>
114   <el-menu
115     default-active="activeIndex"
116     class="el-menu-vertical-demo"
117     :collapse="isCollapse"
118     unique-opened
119     background-color="#304156"

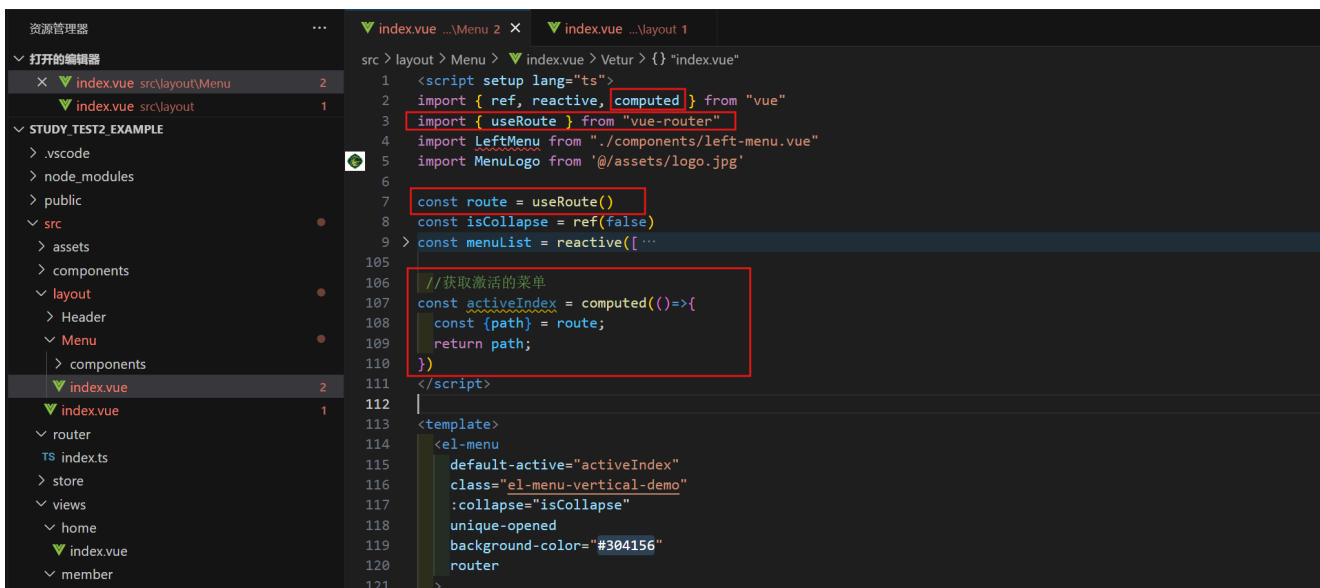
```

```
120     router
121   >
122     <div class="logo">
123       
124       <div class="logo-title">
125         <span>图书借阅</span>
126         <span>管理系统</span>
127       </div>
128     </div>
129     <left-menu :menuList="menuList" />
130   </el-menu>
131 </template>
132
133 <style lang="scss" scoped>
134 .logo {
135   display: flex;
136   width: 100%;
137   height: 80px;
138   line-height: 60px;
139   text-align: center;
140   cursor: pointer;
141   align-items: center;
142   justify-content: center;
143
144   img {
145     width: 36px;
146     height: 36px;
147     margin-right: 12px;
148     margin-left: 12px;
149   }
150
151   .logo-title {
152     color: #fff;
153     font-weight: 800;
154     font-size: 22px;
155     line-height: normal;
156     display: flex;
157     flex-direction: column;
158   }
159 }
160
161 .el-menu-vertical-demo:not(.el-menu--collapse) {
162   min-height: 400px;
163 }
164 .el-menu {
165   height: 100%;
166   border-right: none;
167 }
168 :deep(.el-sub-menu .el-sub-menu__title){
```

```

169     color: #f4f4f5 !important;
170 }
171 :deep(.el-menu .el-menu-item){
172     color: #bfcabd9;
173 }
174 /* 菜单点中文字的颜色 */
175 :deep(.el-menu-item.is-active){
176     color: #409eff !important;
177 }
178 /* 当前打开菜单的所有子菜单颜色 */
179 :deep(.is-opened .el-menu-item){
180     background-color: #1f2d3d !important;
181 }
182 /* 鼠标移动菜单的颜色 */
183 :deep(.el-menu-item:hover){
184     background-color: #001528 !important;
185 }
186 </style>

```



9.菜单收缩

9.1 store 文件夹下新建 modules 文件夹，然后新建 menu.ts

在 store 文件夹下新建 modules 文件夹，然后新建 menu.ts 文件。

```

1  import { defineStore } from 'pinia'
2  import { ref, computed } from 'vue'
3  import type { MenuOptions } from '../interface'
4
5  export const useMenuStore = defineStore('MenuState', () => {
6    // state
7    const isCollapse = ref(false)

```

```

8     const menuList = ref<MenuOptions[]>([])
9     const activeMenu = ref('learning-square')
10    const closeMenuList = ref<string[]>([])
11
12    // getters
13    const getActiveMenu = computed(() => activeMenu.value)
14    const getCloseMenuList = computed(() => closeMenuList.value)
15    const getMenuList = computed(() => menuList.value)
16
17    // actions
18    function setCollapse() {
19        isCollapse.value = !isCollapse.value
20    }
21
22    function setMenuList(list: MenuOptions[]) {
23        menuList.value = list
24    }
25
26    function setActiveMenu(menu: string) {
27        activeMenu.value = menu
28    }
29
30    function setCloseMenuList(list: string[]) {
31        closeMenuList.value = list
32    }
33
34    return {
35        // state
36        isCollapse,
37        menuList,
38        activeMenu,
39        closeMenuList,
40        // getters
41        getActiveMenu,
42        getCloseMenuList,
43        getMenuList,
44        // actions
45        setCollapse,
46        setMenuList,
47        setActiveMenu,
48        setCloseMenuList,
49    }
50 }, {
51     // persist: {
52     //     paths: ['activeMenu'],
53     // },
54     persist: true,
55 })

```

在 `store` 文件夹下新建 `interface` 文件夹，然后新建 `index.ts` 文件。

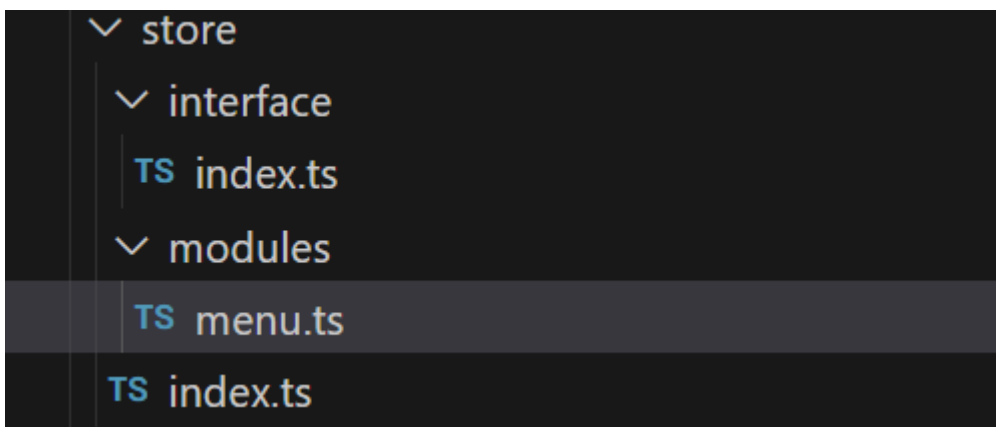
```
1  interface Meta {
2    permission?: string
3    icon?: string
4    title: string
5  }
6
7  interface Child {
8    permission: string
9    path: string
10   name: string
11   component: string
12   icon: string
13   meta: {
14     title: string
15   }
16 }
17
18 export interface MenuOptions {
19   path?: string
20   meta: Meta
21   name: string
22   top?: number
23   icon: string
24   children?: Child[]
25 }
26
27 export interface ThemeConfigProp {
28   primary: string
29   isGrey: boolean
30   isWeak: boolean
31 }
32
33 /* GlobalState */
34 export interface GlobalState {
35   token: string
36   userInfo: any
37   language: string
38   themeConfig: ThemeConfigProp
39 }
40
41 /* MenuState */
42 export interface MenuState {
43   isCollapse: boolean
44   menuList: MenuOptions[]
45   activeMenu: string
46   closeMenuList: string[]
47 }
```

```

48
49  /* TabsState */
50  export interface TabsState {
51    tabsMenuValue: string
52    tabsMenuList: MenuOptions[]
53  }
54
55  /* AuthState */
56  export interface AuthState {
57    authButtons: {
58      [propName: string]: any
59    }
60    authRouter: string[]
61  }

```

目录结构如下。



9.2 引入 pinia-plugin-persistedstate 插件

若 9.1 报错，则下载 pinia-plugin-persistedstate 插件。

```
1  npm add pinia-plugin-persistedstate
```

并修改 main.ts 文件。

```

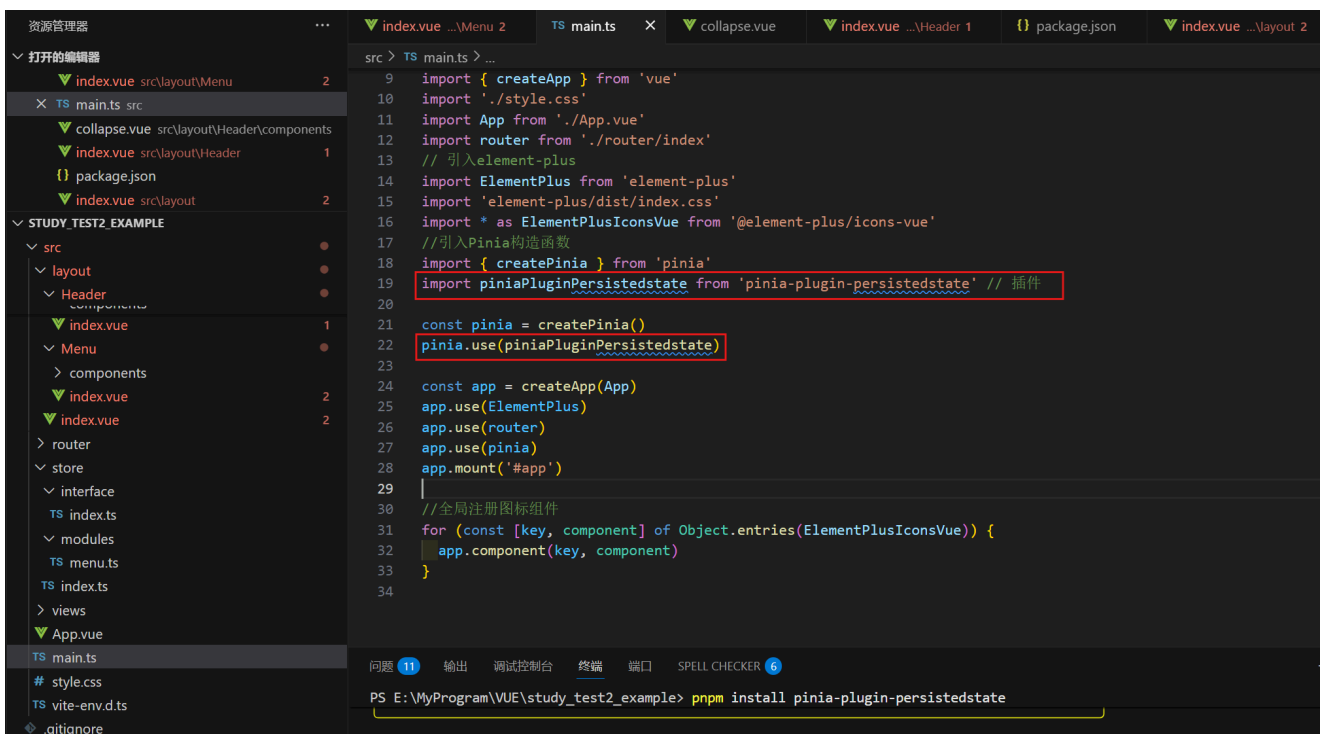
1  import { createApp } from 'vue'
2  import './style.css'
3  import App from './App.vue'
4  import router from './router/index'
5  // 引入element-plus
6  import ElementPlus from 'element-plus'
7  import 'element-plus/dist/index.css'
8  import * as ElementPlusIconsVue from '@element-plus/icons-vue'
9  //引入Pinia构造函数
10 import { createPinia } from 'pinia'
11 import piniaPluginPersistedstate from 'pinia-plugin-persistedstate' // 插
    件

```

```

12
13 const pinia = createPinia()
14 pinia.use(piniaPluginPersistedstate)
15
16 const app = createApp(App)
17 app.use(ElementPlus)
18 app.use(router)
19 app.use(pinia)
20 app.mount('#app')
21
22 //全局注册图标组件
23 for (const [key, component] of Object.entries(ElementPlusIconsVue)) {
24   app.component(key, component)
25 }

```



9.3 在 layout/Header 下新建 collapse.vue 组件

```

1 <script setup lang="ts">
2 import { computed } from "vue";
3 import { Fold, Expand } from "@element-plus/icons-vue";
4 import { useMenuStore } from "@store/modules/menu";
5
6 // 获取store
7 const store = useMenuStore();
8 // 获取collapse状态
9 const status = computed(() => {
10   return !store.isCollapse; // 直接访问 isCollapse 状态
11 });
12 // 改变collapse状态
13 const setCollapse = () => {

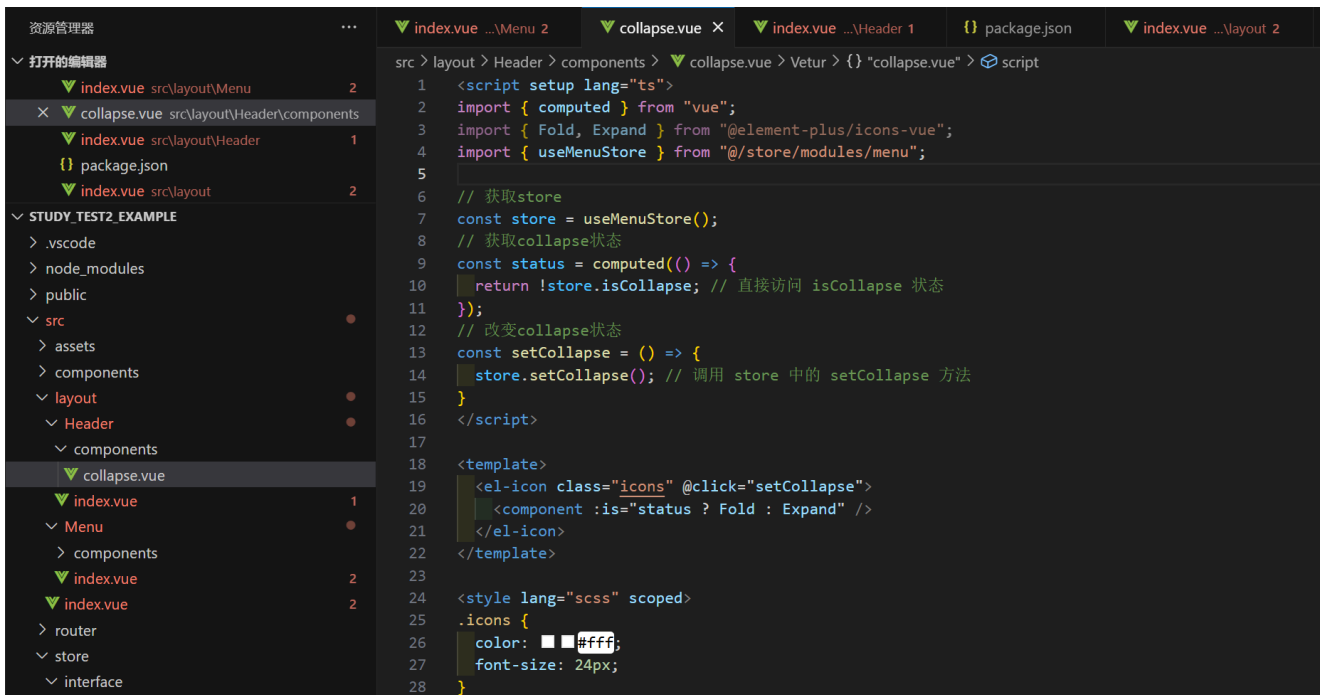
```

```

14     store.setCollapse(); // 调用 store 中的 setCollapse 方法
15 }
16 </script>
17
18 <template>
19     <el-icon class="icons" @click="setCollapse">
20         <component :is="status ? Fold : Expand" />
21     </el-icon>
22 </template>
23
24 <style lang="scss" scoped>
25 .icons {
26     color: #fff;
27     font-size: 24px;
28 }
29 </style>

```

目录结构及代码如下。

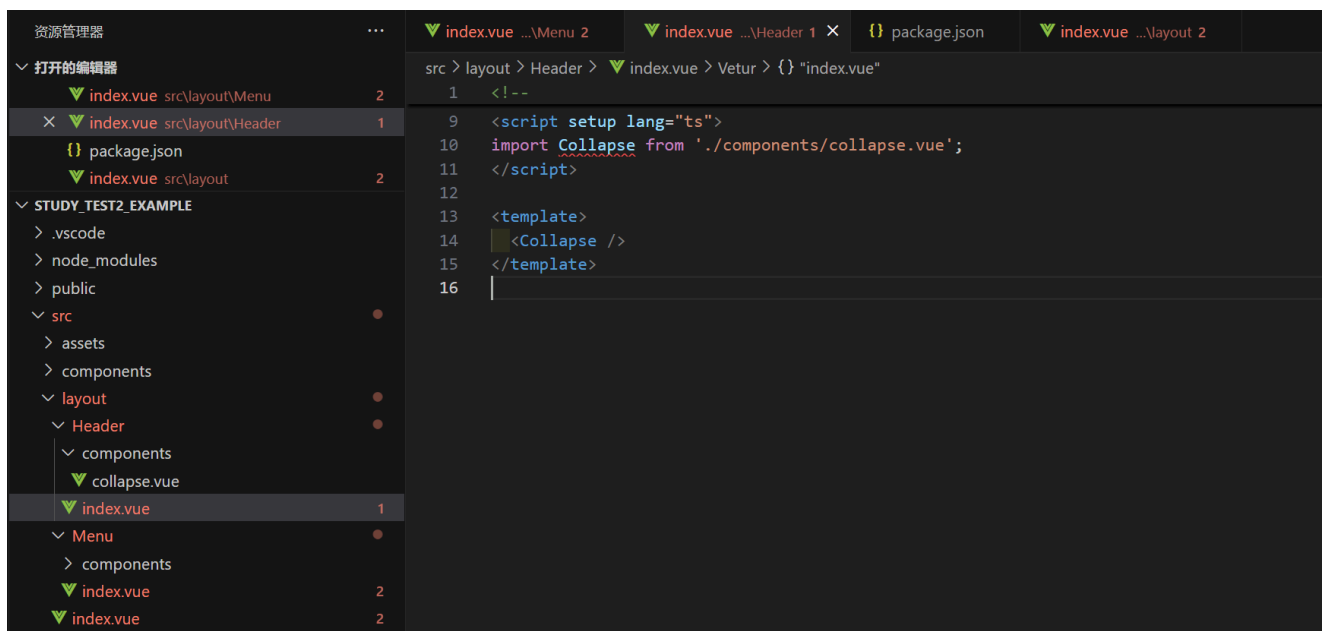


9.4 修改 Header 文件夹中的 index.ts 组件

```

1  <script setup lang="ts">
2  import Collapse from './components/collapse.vue';
3  </script>
4
5  <template>
6      <Collapse />
7  </template>

```



9.5 修改 layout\index.vue 的 header 样式

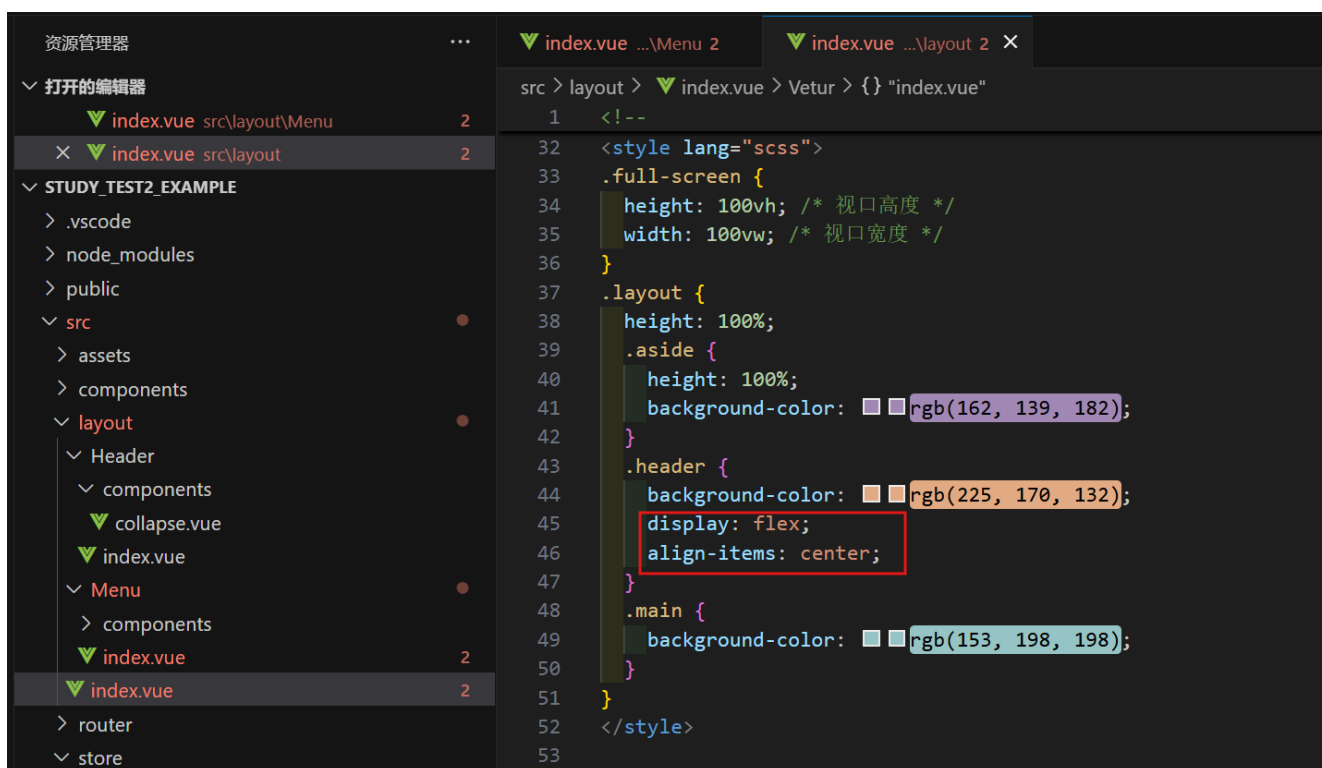
删除原有 width，添加布局。

```
1 <script lang="ts" setup>  
2 import CustomHeader from './Header/index.vue'  
3 import CustomMenu from './Menu/index.vue'  
4 </script>  
5  
6 <template>  
7   <div class="full-screen">  
8     <el-container class="layout">  
9       <el-aside width="auto" class="aside">  
10        <custom-menu />  
11      </el-aside>  
12      <el-container>  
13        <el-header class="header">  
14          <custom-header />  
15        </el-header>  
16        <el-main class="main">  
17          <router-view />  
18        </el-main>  
19      </el-container>  
20    </el-container>  
21  </div>  
22 </template>  
23  
24 <style lang="scss">  
25 .full-screen {  
26   height: 100vh; /* 视口高度 */  
27   width: 100vw; /* 视口宽度 */  
28 }  
29 .layout {
```

```

30     height: 100%;
31     .aside {
32         height: 100%;
33         background-color: rgb(162, 139, 182);
34     }
35     .header {
36         background-color: rgb(225, 170, 132);
37         display: flex;
38         align-items: center;
39     }
40     .main {
41         background-color: rgb(153, 198, 198);
42     }
43 }
44 </style>

```



9.6 修改 layout\Menu\index.vue 组件

```

1  <script setup lang="ts">
2  import { reactive, computed } from "vue"
3  import { useRoute } from "vue-router"
4  import { useMenuStore } from "@store/modules/menu"
5  import LeftMenu from "../components/left-menu.vue"
6  import MenuLogo from '@assets/logo.jpg'
7
8  const route = useRoute()
9  const colstore = useMenuStore();
10 const isCollapse = computed(() => {
11     return !colstore.isCollapse;

```

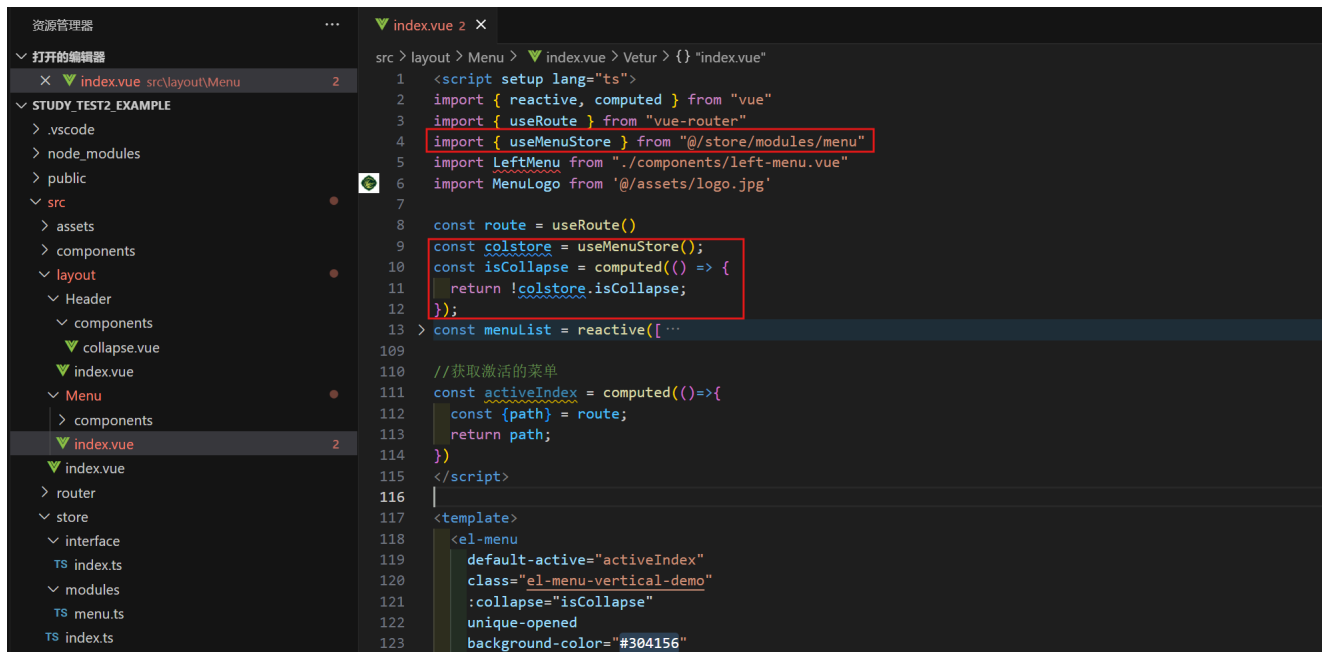
```
12 });
13 const menuList = reactive([
14   {
15     path: "/home",
16     component: "Layout",
17     name: "home",
18     meta: {
19       title: "首页",
20       icon: "House",
21       roles: ["sys:home"],
22     },
23   },
24   {
25     path: "/system",
26     component: "Layout",
27     name: "system",
28     meta: {
29       title: "系统管理",
30       icon: "Setting",
31       roles: ["sys:manage"],
32     },
33     children: [
34       {
35         path: "/userList",
36         component: "/system/User/UserList",
37         name: "userList",
38         meta: {
39           title: "员工管理",
40           icon: "UserFilled",
41           roles: ["sys:user"],
42         },
43       },
44       {
45         path: "/roleList",
46         component: "/system/Role/RoleList",
47         name: "roleList",
48         meta: {
49           title: "角色管理",
50           icon: "Wallet",
51           roles: ["sys:role"],
52         },
53       },
54       {
55         path: "/menuList",
56         component: "/system/Menu/MenuList",
57         name: "menuList",
58         meta: {
59           title: "菜单管理",
60           icon: "Menu",
```

```
61         roles: ["sys:menu"],
62     },
63 },
64 ],
65 },
66 {
67     path: "/memberRoot",
68     component: "Layout",
69     name: "memberRoot",
70     meta: {
71         title: "会员管理",
72         icon: "Setting",
73         roles: ["sys:memberRoot"],
74     },
75     children: [
76         {
77             path: "/cardType",
78             component: "/member/CardType",
79             name: "cardType",
80             meta: {
81                 title: "会员卡类型",
82                 icon: "UserFilled",
83                 roles: ["sys:cardType"],
84             },
85         },
86         {
87             path: "/memberList",
88             component: "/member/MemberList",
89             name: "memberList",
90             meta: {
91                 title: "会员管理",
92                 icon: "Wallet",
93                 roles: ["sys:memberList"],
94             },
95         },
96         {
97             path: "/myFee",
98             component: "/system/FeeList",
99             name: "myFee",
100            meta: {
101                title: "我的充值",
102                icon: "Menu",
103                roles: ["sys:myFee"],
104            },
105        },
106    ],
107 },
108 ])
109
```

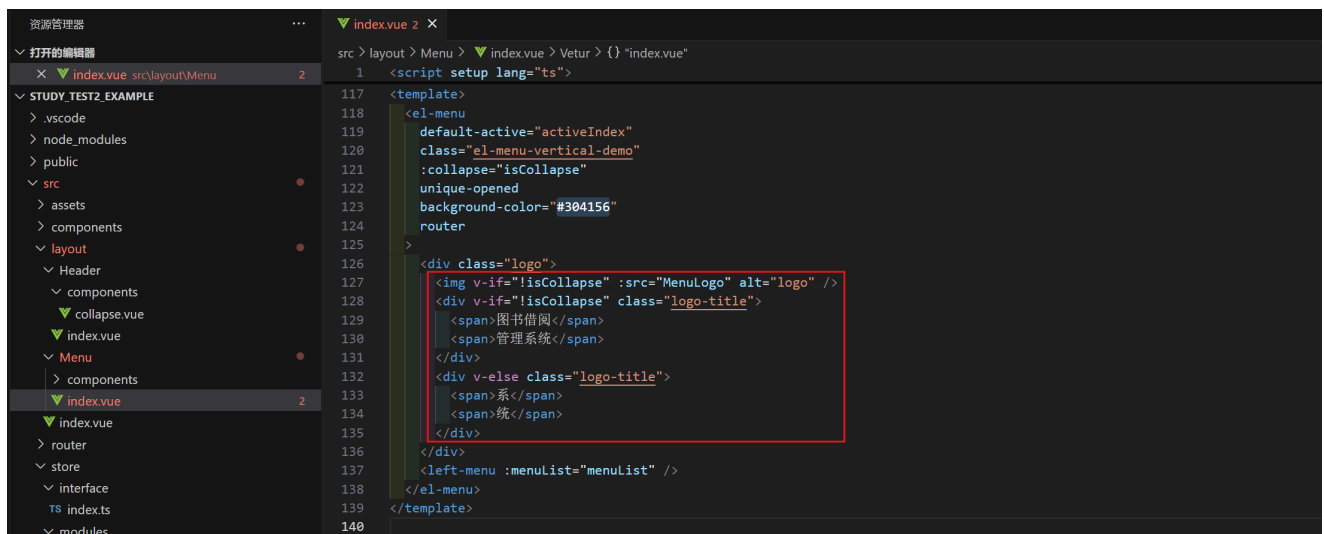
```
110 //获取激活的菜单
111 const activeIndex = computed(()=>{
112     const {path} = route;
113     return path;
114 })
115 </script>
116
117 <template>
118     <el-menu
119         default-active="activeIndex"
120         class="el-menu-vertical-demo"
121         :collapse="isCollapse"
122         unique-opened
123         background-color="#304156"
124         router
125     >
126         <div class="logo">
127             
128             <div v-if="!isCollapse" class="logo-title">
129                 <span>图书借阅</span>
130                 <span>管理系统</span>
131             </div>
132             <div v-else class="logo-title">
133                 <span>系</span>
134                 <span>统</span>
135             </div>
136         </div>
137         <left-menu :menuList="menuList" />
138     </el-menu>
139 </template>
140
141 <style lang="scss" scoped>
142 .logo {
143     display: flex;
144     width: 100%;
145     height: 80px;
146     line-height: 60px;
147     text-align: center;
148     cursor: pointer;
149     align-items: center;
150     justify-content: center;
151
152     img {
153         width: 36px;
154         height: 36px;
155         margin-right: 12px;
156         margin-left: 12px;
157     }
158 }
```

```
159     .logo-title {
160         color: #fff;
161         font-weight: 800;
162         font-size: 22px;
163         line-height: normal;
164         display: flex;
165         flex-direction: column;
166     }
167 }
168
169 .el-menu-vertical-demo:not(.el-menu--collapse) {
170     width: 213px;
171     min-height: 400px;
172 }
173 .el-menu {
174     height: 100%;
175     border-right: none;
176 }
177 :deep(.el-sub-menu .el-sub-menu__title){
178     color: #f4f4f5 !important;
179 }
180 :deep(.el-menu .el-menu-item){
181     color: #bfcabd9;
182 }
183 /* 菜单点中文字的颜色 */
184 :deep(.el-menu-item.is-active){
185     color: #409eff !important;
186 }
187 /* 当前打开菜单的所有子菜单颜色 */
188 :deep(.is-opened .el-menu-item){
189     background-color: #1f2d3d !important;
190 }
191 /* 鼠标移动菜单的颜色 */
192 :deep(.el-menu-item:hover){
193     background-color: #001528 !important;
194 }
195 </style>
```

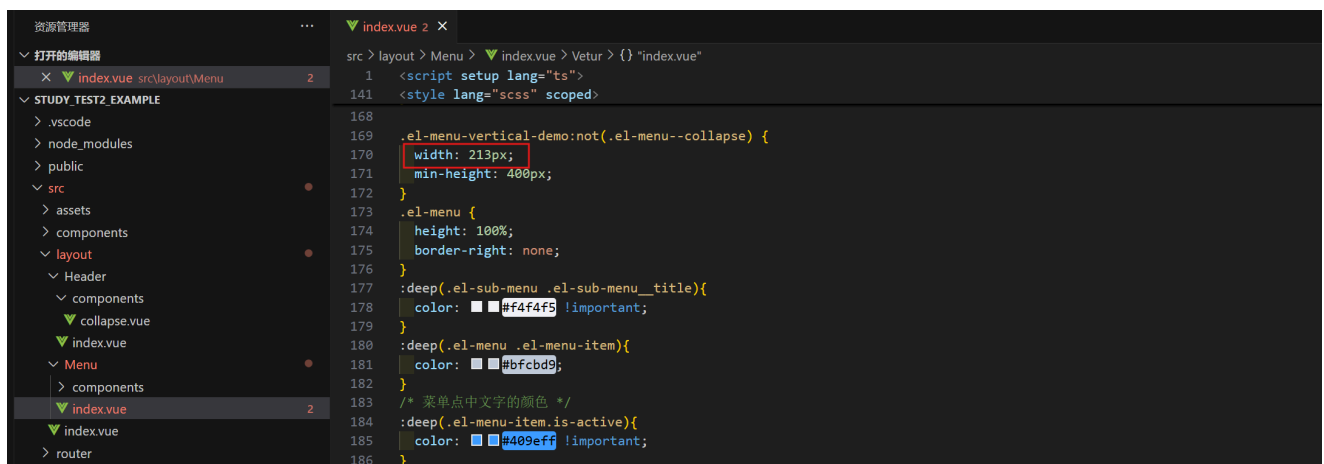
添加 `:collapse` 属性。



修改 logo 及标题的缩放属性。

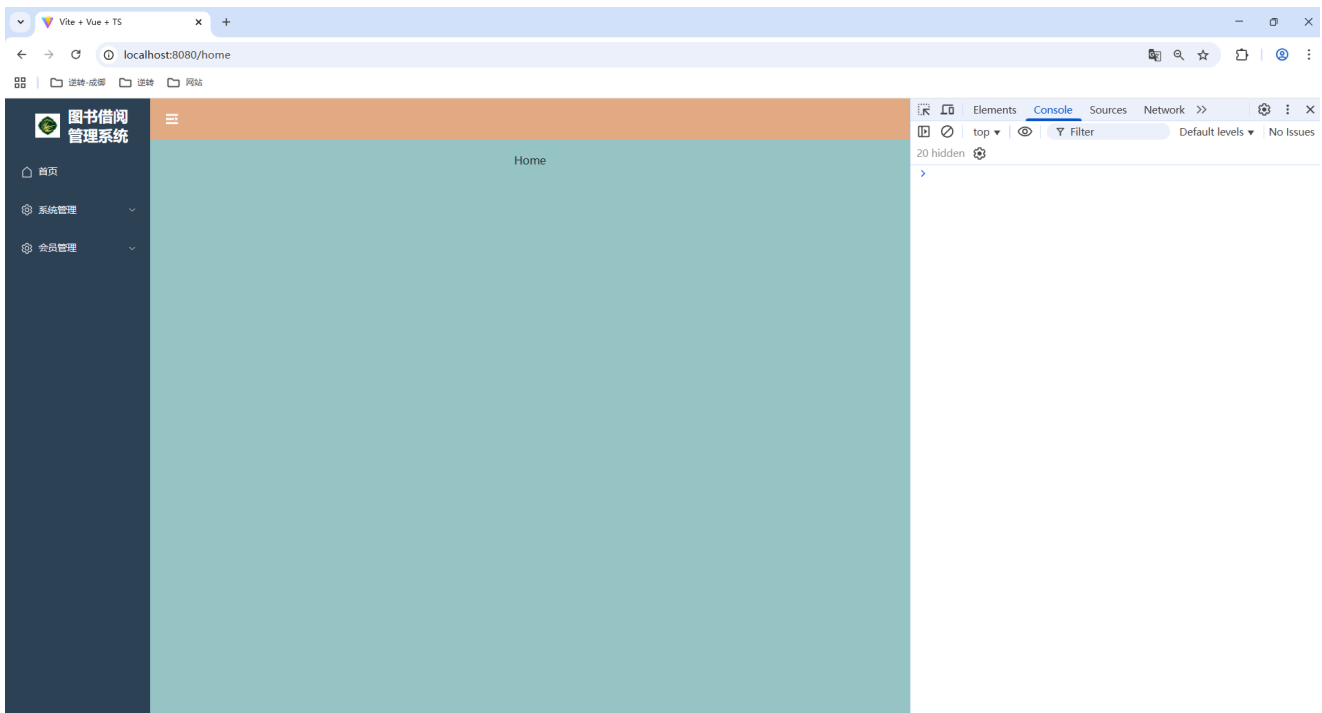


添加目录展开时的宽度。

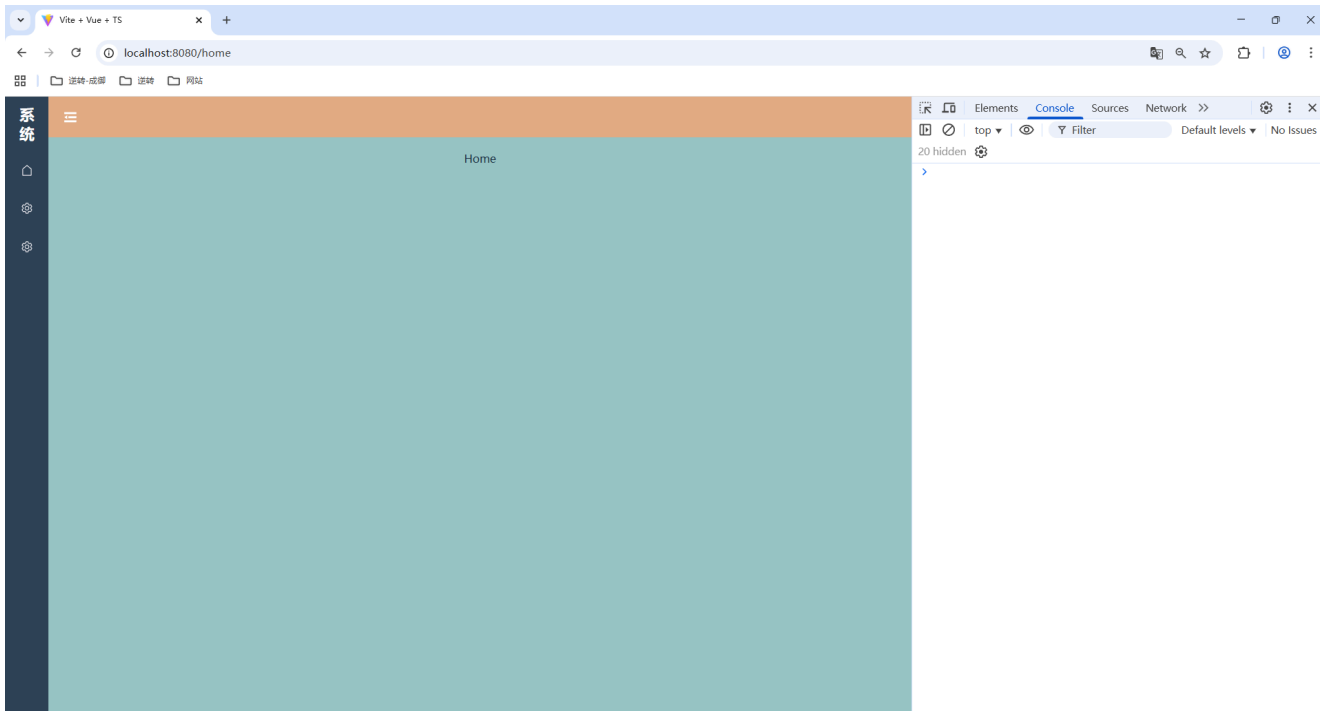


9.7 成果展示

目录展开。



目录收拢。



10.面包屑导航制作

参考官方文档：[Element Plus\]\(https://element-plus.org/zh-CN/component/breadcrumb\)](https://element-plus.org/zh-CN/component/breadcrumb)

10.1 新建 `layout\Header\components\bread-crumb.vue` 文件